

Patient Evaluation and Problem-Oriented Treatment Planning

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Excellence in dental care is achieved through the dentist's ability to assess the patient, determine his or her needs, design an appropriate plan of treatment, and execute the plan with proficiency. Inadequately planned and sequenced treatment, even when well executed, will result in less-than-ideal care. The process of identifying problems and designing the treatment for those problems is the essence of treatment planning and the focus of this chapter.

As an integral part of comprehensive dental care, treatment planning for the restoration of individual teeth must be done in concert with the diagnosis of problems and treatment planning for the entire masticatory system. The objective of this chapter is to present a problem-oriented approach to treatment planning for restorative dentistry. This approach begins with a comprehensive patient evaluation and gradually narrows its focus to the restoration of individual teeth. Emphasis is placed on the decision-making processes involved in identifying problems related to restorative dentistry and determining their causes, which will affect the treatment

outcome; assessing the demands of the oral environment; and selecting the materials, operative modalities, and sequence best suited to the treatment of these problems.

The Problem-Oriented Treatment Planning Model

Treatment planning is generally accomplished with either a treatment-oriented model or a problem-oriented model. In the *treatment-oriented model*, the dentist examining the patient finds certain intraoral conditions and mentally equates those problems to the need for certain forms of treatment. The examination findings are summarized in the form of a list of needed treatments, which then becomes the treatment plan. The *problem-oriented model* requires that the examination lead to the formulation of a list of problems. Each problem on the list is then considered in terms of treatment options, each of which has different advantages and disadvantages. The optimal solution for each problem is then chosen, and, after sequencing, this list of solutions becomes the treatment plan.

For patients with only a few, uncomplicated problems, the outcomes are similar whether the treatment plan is problem based or treatment based. In more complex cases, problems are often interrelated, and the solution to one problem may affect the treatment needed to resolve other problems. In these instances, the process of identifying and listing the individual problems enables the dentist to think through each one and to consider the various options for treating it without getting lost in the magnitude of the overall task.

The problem-oriented approach directs the dentist to perform a systematic evaluation of the patient so that no problems are overlooked, either in diagnosis or in treatment planning. It is designed to prevent tunnel vision of obvious pathoses at the expense of less obvious but equally important problems.

The problem-oriented treatment planning process includes the following steps: a thorough evaluation of the patient's general health and the stomatognathic system; identification of the problems requiring treatment; and development of an integrated treatment plan. Armed with knowledge of all the problems, the clinician envisions the state of the dentition after the seriously compromised and nonrestorable teeth have been removed. Based on this vision of the remaining sound dentition, the dentist visualizes the optimal state to which the patient's dentition can be restored and maintained. Treatment needed to achieve this optimal result can then be documented. This process of planning in reverse, starting with the desired end result, often enables the clinician to identify previously unrecognized or unforeseen problems and add them to the existing problem list to ensure that treatment of each

problem on the problem list is consistent with the desired optimal treatment goal. A list of integrated treatment steps can then be generated.

Problem list formulation

The dentist initially evaluates the patient from a subjective standpoint, ascertaining the chief complaint and the patient's goals of treatment. A medical and a dental history are then elicited. The objective portion of the assessment consists of a categorical evaluation of the patient, beginning with vital signs and an extraoral head and neck examination and progressing through a thorough intraoral evaluation. The examination procedures are standardized and routinely accomplished in the same order and fashion to simplify the procedure and to ensure that crucial steps are not omitted. Related nonclinical portions of the evaluation include examinations of radiographs, diagnostic casts (usually mounted on a semiadjustable articulator), and photographs.

The objectives of the examination are to distinguish normal from abnormal findings and to determine which of the abnormal findings constitute problems that require treatment or will influence treatment. From the findings of the initial examination, a problem list is established. If the problems are listed under categorical headings (eg, periodontal problems, endodontic problems), the dentist is unlikely to omit problems. This list is dynamic and can be modified as new problems arise.

Problem-oriented planning

In the next phase of treatment planning, the dentist considers the various problems with which the patient presents and uses clinical judgment based on evidence-based treatment concepts to estimate which teeth have a sufficiently favorable prognosis to justify being retained and which teeth, if any, should be removed. The dentist visualizes the state of the dentition after the removal of nonsalvageable teeth and then forms a mental image of the optimal condition to which the patient can be rehabilitated. This visualization requires the dentist to decide which teeth need to be replaced and which form of prosthodontic replacement and restorative treatment is most appropriate. Once this optimal condition has been visualized, a treatment solution is proposed for each problem on the problem list, with each individual solution planned to coincide with the final visualized optimal treatment objective.

If the treatment plan for any of the individual problems conflicts with the optimal treatment plan, either the treatment for the individual problem or the optimal

treatment goal must be altered until they are coincident. When the clinician believes that, in consideration of all the problems and proposed treatments, the optimal treatment objective is feasible and maintainable, this list of individual treatments becomes the unsequenced treatment plan. For example, a patient presents with a chief complaint of a broken right maxillary central incisor that will require a crown to restore it, but the maxillary anterior facial gingival contours are asymmetric, which adversely affects the patient's smile. Esthetic crown lengthening of these teeth would greatly improve the smile. If the patient agrees to this procedure, then the broken incisor should initially be restored with a long-term provisional crown to satisfy the chief complaint. The esthetic crown lengthening would then be performed and a definitive crown placed following a satisfactory healing period.

Treatment sequencing

The final step in treatment planning, sequencing the treatment, is completed by arranging the solutions to the various problems in a set order (Box 2-1). The proposed treatment sequencing follows the logic of the medical model, so disease is treated in the priority of importance to the patient's overall health. This method of sequencing ignores the common technique of treating by specialty, where, for example, all the periodontal care is provided, followed by the endodontic care, which is followed by the restorative care.

Box 2-1 Sequence of treatment

- Chief complaint
- Medical/systemic care
- Emergency care
- Treatment plan presentation
- Disease control
- Reevaluation
- Definitive care
- Maintenance caren

Chief complaint

The patient's chief complaint should be addressed at the outset of treatment, even if only via discussion, and even if definitive treatment of this problem will be deferred.

Medical/systemic care

The medical/systemic care phase includes aspects of treatment that affect the patient's systemic health. These take precedence over the treatment of dental problems and must be considered before dental problems are addressed. This most commonly includes medically related diagnostic tests and consultations. An example is the investigation of the status and control of a patient's hypertension, anticoagulant therapy, or diabetes.

Emergency care

Problems addressed in the emergency care phase include those involving head and neck pain or infection. They are treated before routine dental problems but after acute problems involving the patient's systemic health. Evidence-based clinical judgment is exercised to determine the relative importance of systemic problems and dental emergency problems. A review of this topic is found in the text by Little et al.[1]

Treatment plan presentation

The treatment plan presentation (and patient acceptance of the treatment plan) should precede all nonemergency dental care. Presentation and discussion of the proposed treatment are the basis of informed consent and must not be overlooked. The use of intraoral and extraoral photographs of the patient often provides valuable information for educating the patient during this presentation. In addition to the primary or optimal treatment plan, the dentist should be prepared to present alternative plans that may be indicated based on extenuating circumstances, such as patient finances or the therapeutic response of teeth crucial to the success of the plan.

Disease control

The disease control phase consists of treatment designed to arrest active disease. Examples include endodontic treatment to control infection; periodontal treatment to control inflammation; and restorative care, linked with behavior modification, to control caries. Treatment in this phase is aimed at the control of active disease so that the disease processes would not progress even if no treatment beyond disease control were provided.

Reevaluation

The reevaluation phase consists of a formal reassessment, during which the dentist decides if all factors, including such criteria as the patient's treatment goals, oral hygiene, behavior modification, and response to periodontal therapy, warrant continuing with the original treatment plan. This is an important phase of treatment because it provides a predetermined point at which both patient and clinician may elect to alter or even discontinue treatment.

Definitive care

The definitive care phase is the final phase of treatment preceding maintenance care. Many of the procedures accomplished within the disease control phase, such as removal of carious tooth structure and placement of direct restorations, achieve both disease control and definitive restoration; however, a number of procedures that go beyond the treatment of active disease are possible. These include procedures designed to enhance function and esthetics, such as orthodontics, surgery, prosthodontics, and cosmetic restorative procedures. Treatment sequencing for most of these modalities is beyond the scope of this text; a detailed and comprehensive review is provided by Stefanac and Nesbit.[2]

Maintenance care

Maintenance care is an ongoing phase designed to maintain the results of the previous treatment and prevent recurrence of disease. The maintenance phase generally focuses on the maintenance of periodontal health; the prevention, detection, and treatment of caries; and the prevention of dental attrition, erosion, and abrasion.

Dental History and Chief Complaint

The key to successful treatment planning lies in identifying the problems that are present and formulating a treatment plan that addresses each problem so that each phase of treatment is designed to lead to the final, optimal treatment goal. The dentist who follows this approach begins by listening carefully to the patient and asking relevant questions. A thorough dental history serves as a guide for the clinical examination.

The dental history is divided into three components: *(1)* chief complaint, *(2)* dental treatment, and *(3)* symptoms related to the stomatognathic system. The chief complaint is addressed first and is recorded in the dental record in the patient's own words. By discussing the patient's chief concern at the outset, the dentist

accomplishes two important goals. First, the patient feels that his or her problems have been recognized, and the doctor-patient relationship begins positively; second, by writing out the chief complaint, the dentist ensures that it will not be omitted from the problem list. It is not uncommon to encounter a patient who has a multitude of significant dental problems but only a minor chief complaint. If the dentist focuses too quickly on the other problems and omits a discussion of the chief complaint, the patient may question the dentist's ability and desire to resolve the patient's chief concern.

A brief history of past dental treatment can provide useful information. The number and frequency of past dental visits reflect the patient's dental awareness and the priority he or she places on oral health. The dentist should elicit information about the past treatment of specific problems, as well as the patient's tolerance for dental treatment. All of this information can be of use in developing the treatment plan.

Questions about previous episodes of fractured or lost restorations, trauma, infection, sensitivity, and pain can elicit information that will alert the dentist to possible problems and guide him or her during the clinical and radiographic examination. Patients may not volunteer this information; hence, specific questions regarding thermal sensitivity, discomfort during chewing, gingival bleeding, and pain are warranted. When there is a history of symptoms indicative of pulpal damage or incomplete tooth fracture, specific diagnostic tests should be performed during the clinical examination.

Clinical Examination

Every patient examination should include an assessment of the extraoral and intraoral soft tissue conditions as well as the dentition. The reader is referred to standard texts on physical examination for details. The temporomandibular joint and associated musculature should be evaluated and a systematic assessment of any signs or symptoms of pathology conducted.[3] For the purpose of restorative treatment planning, the intraoral assessment involves an examination of the periodontium, dentition, and occlusion. Specific diagnostic tests may be performed as indicated, and a radiographic examination is completed at this time. The dentist should be sure to complete one portion of the evaluation before beginning another aspect of the examination (Box 2-2). The findings from each area are placed under the appropriate heading in the problem list. Some problems may be noted in the

evaluation of more than one system. For example, gingival bleeding and periodontal inflammation resulting from the impingement of a restoration on the periodontal attachment would be noted in both the periodontal examination and the evaluation of the existing restorations. At this stage, such duplication of effort is acceptable in the interest of completeness.

	Elements of the clinical examination
I. Evaluation of the dentition • Caries risk and plaque • Caries lesion detection and disease activity • Pulp • Existing restorations • Occlusion and occlusal contours • Axial tooth surfaces • Tooth integrity and fractures • Esthetics	II. Evaluation of the periodontium • Disease activity • Structure and contour of bony support • Mucogingival assessment • Tooth mobility III. Evaluation of radiographs VI. Evaluation of diagnostic casts (may include a diagnostic mounting in centric relation and use of a facebow)

The following sections describe the elements of the intraoral examination used to establish the restorative dentistry problem list.

Evaluation of the dentition

Caries risk and plaque

An assessment of caries risk (see chapter 5) should be accomplished, and the presence of plaque should be documented with a standardized plaque index. The O'Leary index, for example, is a simple, effective measure of plaque accumulation.[4] The use of a standardized index permits an objective assessment of plaque accumulation. Depending on initial findings, a dietary analysis and/or unstimulated salivary flow rate determination may be indicated. The determination of baseline caries risk and plaque levels at the time of initial examination provides a basis for communication with the patient and other clinicians and permits assessment of changes over time. This is important information in establishing a prognosis for restorative care and provides criteria for deciding whether treatment should progress beyond the disease control phase into the definitive rehabilitation stage.

The levels and location of plaque should be established at the outset of the examination. At the conclusion of the examination, the patient may be given a toothbrush and floss and asked to clean the teeth as well as possible. Reassessment

immediately after the cleaning will establish the patient's hygiene ability and reveal the nature of hygiene instructions needed. A patient who sincerely tries to remove plaque but is unsuccessful in certain areas requires instruction in technique, whereas the patient who demonstrates effective hygiene while in the office but consistently presents with high plaque levels has a problem with motivation. This information is important in designing the treatment plan: A plan requiring a great deal of patient participation and compliance would not be appropriate for a patient with inadequate motivation, while a motivated patient who is teachable may well be suited to such a plan.

One of the most reliable indicators of future caries activity is the presence of an existing or recently treated caries lesion.[5] Three additional factors that heighten the risk of caries are *(1)* a large number of cariogenic bacteria, *(2)* frequent ingestion of cariogenic sugars, and *(3)* an inadequate flow of saliva.[6] Patients demonstrating active caries should receive an evaluation that entails more than simply a determination of levels and location of plaque.[7] Both a diet survey and a salivary flow assessment are useful in determining the patient's susceptibility to caries and the caries-related prognosis for restorative treatment.

Diet has been shown to be one of the most significant factors in caries risk. A review of more than 100 studies by van Palenstein Helderman et al[8] demonstrated that the frequency and duration of refined carbohydrate exposure is more predictive of caries occurrence than *Streptococcus mutans* counts. Using a diet survey, the patient itemizes all food and drink intake for a specified period (generally 1 week). From this diary, the dentist can identify the contribution of specific dietary habits to the patient's caries risk and can direct the patient's attention to these areas. The identification and management of episodic sugar and carbohydrate intake (snacking), as well as overall carbohydrate consumption, should be the focus of dietary intervention.[9,10]

Xerostomia, or dry mouth, is associated with increases in the number of cariogenic bacteria and increased caries activity.[11] Saliva provides lubrication; promotes oral clearance of fermentable carbohydrates, sugars, and acids; and possesses antimicrobial components and buffering agents.[11–13] Saliva is critical to tooth remineralization because it is a source of calcium, phosphates, and proline-rich proteins active in recrystallization of the tooth surface.[12,13] The patient's health status, medications, iatrogenic changes, and possibly aging can alter salivary flow and composition and, thereby, caries-risk status. Consideration should be given to assessing salivary flow rates in all caries-active patients. There is not complete

agreement as to the minimum salivary flow rate necessary to maintain oral health. Some authorities suggest that unstimulated salivary flow in a range less than 0.1 to 0.2 mL/min is the criterion for hypofunction.[14–16] However, others correlate clinical symptoms of hypofunction, such as dysphasia, dysphagia, xerostomia, and a higher incidence of caries lesions and/or candidiasis, with an unstimulated flow rate of less than 0.2 mL/min.[17,18]

Because the character of the microflora determines the cariogenicity of the plaque, periodic assessment of the number of cariogenic bacteria present in the plaque can indicate alterations in the caries susceptibility of patients at high risk of developing caries.[19,20] Although higher levels of *S mutans* (greater than 106 colony-forming units [CFUs]/mL of saliva) are not consistently indicative of caries activity, estimates of the number of bacteria are more useful for predicting the absence, rather than the presence, of an active infection.[20,21] By monitoring the levels at baseline and over time, the dentist can assess the effectiveness of caries-management measures.

Once plaque assessments have been completed, an examination of other areas can be accomplished. The visual examination of the dentition should be conducted in a dry field, with adequate lighting, using a mirror and explorer. Ideally, the dentist will employ some form of magnification to aid in the examination. A number of products providing 2× to 6× magnification are commercially available. Some magnifying lenses attach to eyeglasses and can be removed, while others are built directly into the lenses of specially constructed eyeglasses (see Fig 7-68). The use of magnification with adequate lighting significantly enhances the ability of the clinician to detect subtle signs of disease. If the presence of plaque and calculus partially obscures the dentition, debridement is required to accomplish a thorough examination.

Detection of caries lesions

The terms *carious lesion* and *caries lesion* are both acceptable to describe the effect of the caries process on a tooth, and both terms will be used interchangeably throughout this textbook.

Caries lesions may be classified by location into two broad categories: pit and fissure caries lesions and smooth-surface caries lesions (including those involving proximal surfaces, root surfaces, and lesions on other smooth surfaces). Detection of caries lesions requires both clinical (visual and tactile) and radiographic examinations.

Pit and fissure caries lesions. Pit and fissure caries lesions are generally found in areas of incomplete enamel coalescence. These areas are most commonly found on the occlusal surfaces of posterior teeth, the lingual surfaces of maxillary anterior teeth and maxillary molars, and the buccal pits of mandibular molars. Because pit and fissure lesions may begin in small enamel defects that lie in close approximation to the dentinoenamel junction, they may be difficult to detect. A pit and fissure lesion must be fairly extensive to be detected radiographically; these lesions generally appear as crescent-shaped radiolucencies immediately subjacent to the enamel[22] (Fig 2-1).



Fig 2-1*(a)* Occlusal caries. The shadowing around the stained pits in the second molar indicates the presence of carious dentin at the base of the fissure. *(b)* The caries lesion extends well into dentin. *(c)* The typical pattern of an occlusal caries lesion in cross section.

Historically, tactile examination with firm application of a sharp explorer into the fissure was the clinical technique most commonly used by dentists in the United States to locate pit and fissure lesions.[23] A sticky sensation on removal of the explorer has been the classic sign of pit and fissure caries. Clinical studies, however, have shown this method to be unreliable, producing many false-positive and false-negative diagnoses.[24] In addition, an explorer can cause cavitation in a demineralized pit or fissure, precluding the possibility of remineralization.[24–27]

Visual observation, with magnification, of a clean, dry tooth has been found to be a reliable, nondestructive method of detecting pit and fissure caries lesions,[24–28] which appear as gray or gray-yellow opaque areas that show through the enamel (see Fig 2-1a). However, stain within a fissure is not indicative of carious dentin at the base of the fissure.

Fiber-optic transillumination may be helpful in visualizing pit and fissure and other types of caries lesions. A variety of new technologies are being evaluated for detection of caries lesions. A discussion of these new technologies can be found in chapter 5.

When the presence of pit and fissure lesions is uncertain and the patient will be

available for recall evaluations, a sealant may be placed over the suspected area. Clinical investigation by Mertz-Fairhurst et al[29] indicates that sealed caries lesions do not progress. However, placement of sealants in fissures over known carious dentin cannot be recommended at present, because the risk of sealant loss makes this an injudicious practice. Mertz-Fairhurst et al[29] found the placement of a conservative amalgam or resin composite restoration, followed by the placement of a resin fissure sealant over the margins of the restoration and remaining fissures, to be a predictable and relatively conservative treatment for such lesions. A recent systematic review by Azarpazhooh and Main[30] recommended that sealants be placed on all permanent molars without cavitation as soon after eruption as isolation can be achieved.

Smooth-surface caries lesions. Of the three types of smooth-surface caries lesions, proximal lesions are the most difficult to detect clinically. Proximal caries lesions in posterior teeth are generally inaccessible to both visual and tactile examination and are usually detected radiographically. Proximal lesions in anterior teeth may be detected radiographically or with visual examination using transillumination[28] (Fig 2-2). Root caries lesions located on facial or lingual surfaces of the roots present few diagnostic problems. When root-surface lesions occur proximally, however, they are not readily visible on clinical examination and are generally detected through the radiographic examination (Fig 2-3). Smooth-surface caries lesions occurring on enamel in nonproximal areas are not difficult to detect clinically. These lesions, which are most commonly found in patients with high levels of plaque and a cariogenic diet or deficient salivary flow, occur on the facial and lingual enamel surfaces and are readily accessible during visual and tactile examination.

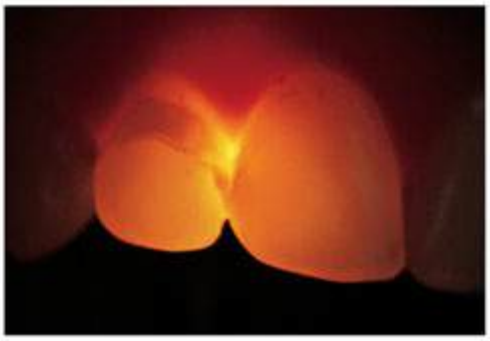


Fig 2-2 Proximal caries lesion detected in an anterior tooth with the use of transillumination.



Fig 2-3

Dental pulp

Evaluation of pulpal vitality in every tooth is not warranted; however, each tooth that will undergo extensive restoration, as well as all teeth that are critical to the plan of treatment and teeth with pulps of questionable vitality, should be tested.

The application of cold is a valuable method of vitality testing. Canned refrigerants present minimal risk to teeth and restorations. A cotton pellet saturated with an aerosol refrigerant spray, such as tetrafluoroethane, is placed on the tooth to determine its vitality. A similar test can be performed by placing a "pencil of ice" (made by freezing water inside a sterilized anesthetic cartridge) against a tooth.

An additional vitality test involves the use of an electric pulp tester. While it can provide information regarding pulp vitality, this test has limitations; it cannot be used in a wet field or on teeth with metallic proximal surface restorations unless measures are taken to insulate adjacent teeth. Furthermore, the numeric scale of the instrument does not reflect the health of the pulp or its prognosis. The electric pulp tester is merely a means of determining whether the tissue within the pulp senses electrical current. A high score may be due to the presence of a partially necrotic pulp or extensive reparative dentin, or it may be the result of poor contact between the tooth and the pulp tester.

When the results of pulp tests are not congruent with the clinical impression, additional tests are indicated. When neither thermal nor electric pulp tests provide a clear picture of pulp vitality and a restoration is indicated, the preparation can be initiated without the use of anesthetic. This is termed a *test cavity*. If pain or sensitivity is elicited when dentin is cut with a bur, pulpal vitality is confirmed. The restoration may then be completed after administration of local anesthetic.

Pulp vitality should be determined prior to restorative treatment. It is professionally embarrassing to discover that a recently restored tooth was nonvital prior to restoration and subsequently became symptomatic, requiring endodontic

treatment and a replacement restoration. It is advantageous to ascertain the pulpal prognosis of a tooth prior to restorative treatment. When pulpal prognosis is uncertain or guarded, it is often best to perform endodontic therapy before extensive restorative treatment. If the endodontic treatment is completed before restorative care, the repair or replacement of a recently completed large restoration may be avoided.

Planning for endodontic treatment and presenting it as part of the original treatment plan is generally more acceptable to the patient than presenting this treatment option after treatment has begun. An added benefit is that the endodontic prognosis can be established before the dentist commits to restorative care. When endodontic therapy is required, the feasibility of completing the endodontic procedures should be determined early in the course of treatment. The more critical the tooth is to the overall success of the treatment, the more important it becomes to complete the necessary endodontic treatment early in the treatment schedule. It is poor planning to rely on a tooth in the treatment plan when that tooth cannot be successfully treated with endodontics. Endodontic diagnosis can be challenging. A thorough discussion of this subject can be found in the text by Hargreaves and Cohen.[31]

When a posterior tooth has received endodontic treatment, placement of a complete-cuspal-coverage restoration is generally indicated to prevent fracture.[32,33] When an anterior tooth has received endodontic treatment, the least invasive form of restoration that satisfies the esthetic and functional needs of the patient is indicated.[34] If sufficient enamel and dentin remain for support, a bonded restoration, such as a resin composite restoration or a ceramic veneer, is preferred. If there is insufficient support for such a restoration after removal of carious tooth structure or defective restorations or following endodontic access preparation, a ceramic or metal-ceramic crown is the restoration of choice. A post is indicated primarily to retain a core in a tooth with extensive loss of coronal tooth structure.[32,35,36] When a post is needed, preparation of a small post space preserves dentin and provides optimal fracture resistance for the tooth[32,37] (see chapter 21).

Existing restorations

In the course of the intraoral examination, the serviceability of existing restorations must be evaluated. The following general criteria are used to evaluate existing restorations: *(1)* structural integrity, *(2)* marginal opening, *(3)* anatomical form, *(4)* restoration-related periodontal health, *(5)* occlusal and interproximal contacts, *(6)*

recurrent caries lesions, and *(7)* esthetics.

Structural integrity. The structural integrity of a restoration should be evaluated to determine whether it is intact or whether portions of the restoration are partially or completely fractured or missing. The presence of a fracture line dictates replacement of the restoration. If voids are present, the dentist must exercise clinical judgment in determining whether their size and location will weaken the restoration and predispose it to further deterioration or recurrent carious involvement (Fig 2-4).

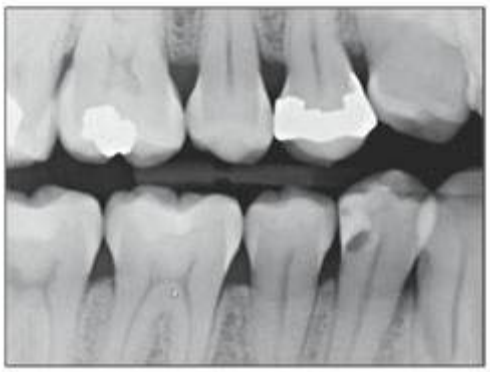


Fig 2-4 Bitewing radiograph illustrating a large void in the Class 2 resin composite restoration of the mandibular right first premolar.

Marginal opening. Few restorations have perfect margins, and the point at which marginal opening dictates replacement of the restoration is difficult to determine. For amalgam restorations, it has been demonstrated that marginal ditching neither implies the presence nor necessarily portends the development of caries lesions[38]; therefore, its existence does not dictate the replacement of amalgam restorations. Because the margins of amalgam restorations become relatively well sealed by the accumulation of corrosion products, a general guideline has been to continue to observe the restoration unless signs of recurrent caries lesions or marginal gaps are present.[39] An accumulation of plaque in the marginal gap is also an indication for repair or replacement of an amalgam restoration. Repair of noncarious marginal gaps in amalgam with a flowable resin composite or a sealant will enhance the longevity of the restoration.[40–42] A long-term, retrospective clinical study by Smales and Hawthorne[43] indicated that the repair of local defects in amalgam restorations is an effective alternative to restoration replacement.

For restorations that do not seal by corrosion, a marginal gap into which the end of a sharp explorer may penetrate should be considered for repair, or the restoration should be replaced. This is especially true for resin composite restorations, because bacterial growth has been shown to progress more readily adjacent to resin

composite than to amalgam or glass-ionomer materials.[44] Repair, sealing, or replacement of resin composite restorations with defective margins will result in significant improvement in the longevity of the restoration.[45] An increased susceptibility to caries has been reported in resin composite restorations whose marginal gaps exceeded 100 to 150 μm.[46] In anterior teeth, replacement is indicated when the tooth structure adjacent to the marginal gap becomes carious or when marginal staining is esthetically unacceptable. Often this stain may be associated with unbonded resin flash that can usually be polished away. The dentist must carefully evaluate the restoration margin to determine the need for replacement (Fig 2-5).



Fig 2-5

The presence of a marginal gap is less critical for restorations with anticariogenic properties (eg, glass-ionomer cement). Both in vitro[47–53] and in vivo[54–58] studies have shown that tooth structure adjacent to glass-ionomer restorations is less susceptible to caries attack than that adjacent to either resin composite or amalgam restorations. Consequently, restorations with anticariogenic properties generally should be replaced not because of marginal ditching but rather when a frank caries lesion has occurred or when some other defect indicates the need for treatment.

Anatomical form. *Anatomical form* refers to the degree to which the restoration duplicates the original contour of the intact tooth. Common problems include overcontouring, undercontouring, uneven or flat marginal ridges, inadequate facial and lingual embrasures, and lack of occlusal or gingival embrasures. Many restorations exhibit one or more of these problems yet adequately serve the needs of the patient and do not require replacement. The critical factor in determining the need for replacement is not whether the contour is ideal but whether pathosis has resulted, or is likely to result, from the poor contour.

Restoration-related periodontal health. Examination of restorations must include an assessment of the effect that existing restorations have on the health of the adjacent periodontium. Problems commonly encountered in this area are *(1)* surface roughness of the restoration, *(2)* interproximal overhangs, and *(3)* impingement of the restoration margin on the zone of attachment, called the *biologic width* or the *dentogingival junction* (the area, approximately 2 mm in the apicocoronal dimension, occupied by the junctional epithelium and the connective tissue attachment) (Fig 2-6; see also Fig 1-32).



Fig 2-6 Bitewing radiograph illustrating an amalgam overhang on the distal of the maxillary first molar.

All three of these phenomena can cause inflammation within the periodontium.[59–61] If restorative material extends vertically or horizontally beyond the cavosurface margin in the region of the periodontal attachment or impinges on the biologic width, the health of the periodontal tissue should be assessed (Fig 2-7). If other local etiologic factors have been removed and periodontal inflammation persists in the presence of these conditions, treatment should be initiated. In the case of overhanging restorations, pathosis may be eliminated and the restoration may be made serviceable simply by removing the overhang. If the periodontal inflammation fails to resolve, the restoration should be replaced. In the case of biologic width impingement, space for a healthy periodontal attachment must be gained through surgical crown lengthening or a combination of orthodontically forced eruption and surgical crown lengthening.



Fig 2-7 Periodontal inflammation caused by the encroachment of crown margins into the periodontal attachment area of the maxillary right central incisor.

Inflammatory changes suggestive of biologic width violations are common on the facial aspects of anterior teeth that have been restored with crowns. On occasion, however, evaluation of the marginal areas reveals inflammation even when an adequate space remains between the crown margin and the periodontal attachment apparatus, leaving the clinician puzzled as to the cause of the problem. If periodontal inflammation persists in the apparent absence of local etiologic factors, including biologic width impingement, the dentist should evaluate the entire cervical circumference of the restoration. Inflammatory changes on the facial aspect of a restoration are sometimes a manifestation of interproximal inflammation. Further evaluation may reveal an interproximal violation of biologic width from which the inflammatory reaction has extended to the more visible facial areas.

Even in the absence of impingement on biologic width, open or rough subgingival margins can harbor sufficient bacterial plaque to generate an inflammatory response. Gingival inflammation around a crown may also be due to an allergic reaction to a material in the crown. Several investigations have shown that low gold content or predominantly base metal alloys, especially those containing nickel and palladium, can cause such reactions, with an incidence of around 7%.[62–66]

During the assessment of existing restorations or the planning of future restorations, the location of margins is an important consideration. Supragingival margins result in significantly less gingival inflammation than do subgingival margins.[67] Supragingival margins should be the goal when overriding concerns (eg, esthetics or requirements for resistance and retention) do not contraindicate their use.

Occlusal and interproximal contacts. The dentist should assess all interproximal contacts with thin dental floss. In addition, the patient should be queried regarding

any problems encountered in the passing of floss through the contacts during home hygiene procedures. Contacts that do not allow the smooth passage of floss must be altered, or the restoration must be replaced, to permit the use of floss. The use of an interproximal contact–smoothing device is often effective in eliminating roughness that impedes the passage of floss (Fig 2-8).



Fig 2-8 An interproximal contact–smoothing device is useful for removing irregularities that impede the passage of floss.

Contacts that are open or excessively light should be evaluated to determine whether pathosis, food impaction, or annoyance to the patient has resulted. When any of these problems is present, steps should be taken to alleviate it. Generally, the placement or replacement of a restoration is required to establish an adequate proximal contact. When an open contact is identified, an attempt should be made to determine its cause. If occlusal contacts have moved a tooth and a restoration is to be placed to close the proximal contact, the occlusal contacts must be altered to prevent the open contact from recurring after the placement of the new restoration.

The occlusal contacts of all restorations should be evaluated to determine if they are serving their masticatory function without creating a symptomatic or pathogenic occlusion. In the absence of periodontally pathogenic bacteria, traumatic occlusion has not been found to initiate loss of periodontal attachment.[68–70] However, in a susceptible host and in the presence of periodontal pathogens, occlusal trauma can play a role in the progression of periodontal disease.[71–74] A more recent study indicates that these previous studies may have underestimated the impact that occlusal trauma has on the periodontium.[75] Existing restorations located in teeth exhibiting significant attachment deficits should be examined closely for the presence of hyperocclusion. Restorations in which occlusal contacts are creating primary occlusal trauma should be altered or replaced, as necessary, to resolve the problem.[76] Restorations that are in significant infraocclusion may permit the supraeruption of teeth and should be considered for replacement.

Recurrent caries lesions. The evaluation for carious tooth structure around existing restorations focuses on an examination of the margins. The dentist must use a combination of visual, tactile, and radiographic examinations to detect the presence of caries lesions. A radiolucent area surrounding a radiopaque restoration on a radiograph or the presence of soft tooth structure generally indicates a caries lesion and warrants either repair or replacement of the restoration.

Discoloration in the marginal areas is a more difficult sign to interpret. It often indicates leakage of some degree. In nonamalgam restorations without anticariogenic properties, discoloration that penetrates the margin often indicates the need for replacement of the restoration (Fig 2-9). This is not a definite indication, however, and clinical judgment is required. For example, in restorations with anticariogenic properties, leakage and staining may be observed with less concern for caries involvement, leaving esthetics as the primary consideration. This is not to imply that restorative materials with caries-resistant properties are immune to caries, however. Caries lesions have been documented adjacent to glass-ionomer restorations.[55,77] If the tooth structure adjacent to the margin of a restoration appears to be carious (either with undermined enamel or cavitation), rather than simply discolored, the restoration should be repaired or replaced, depending on location and extent of caries.

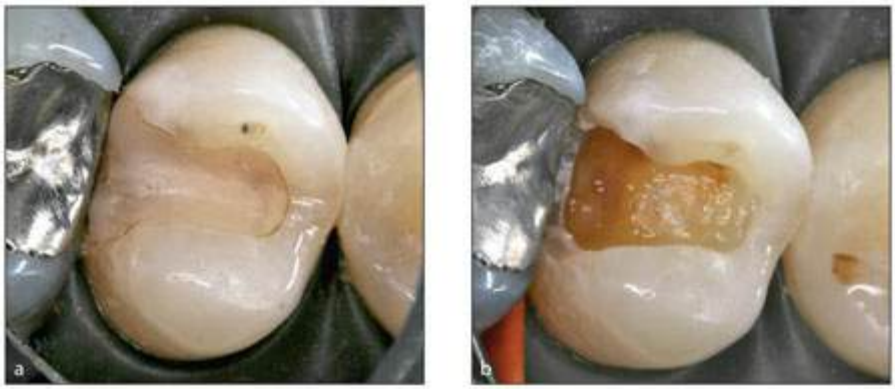


Fig 2-9*(a) (b)*

For amalgam restorations, the decision to replace a restoration when there is discoloration in the adjacent tooth structure is less clear because corrosion products may discolor tooth structure, even in the absence of caries lesions, especially when little dentin is present. When there is no apparent communication between the cavosurface margin and the stained area, and when the discoloration is primarily gray, then metal "show-through" should be suspected and observation is warranted (Fig 2-10). When the discolored area appears yellow or brown and appears to communicate with the cavosurface margin, replacement of the restoration is

indicated (Fig 2-11).



Fig 2-10 The shadow in the mesiofacial aspect of the maxillary right first molar is caused by amalgam that shows through the translucent enamel. No caries lesion is present.



Fig 2-11 The shadow located on the mesiolingual cusp adjacent to the larger occlusal amalgam restoration on the maxillary right first molar indicates the presence of carious dentin.

Esthetics. The esthetic evaluation of existing restorations is highly subjective. When the functional aspects of a restoration are adequate, it is often best to simply inquire whether or not the patient is satisfied with the esthetic appearance of the existing restorations. If the patient expresses dissatisfaction with the appearance of a restoration, the dentist must determine whether improvement is feasible. Care should be taken to ascertain the reason that the original restoration has less than optimal esthetics. An underlying problem may preclude improvement of the original esthetic problem, and an equally unsatisfactory result may occur in the replacement restoration. When replacing a restoration for esthetic reasons only, the dentist must carefully explain the risks (eg, endodontic complications) incurred in replacement.

Some of the more common esthetic problems found in existing restorations are *(1)* display of metal, *(2)* discoloration or poor shade match in tooth-colored restorations, *(3)* poor contour in tooth-colored restorations, and *(4)* poor periodontal tissue response in anterior restorations. (See chapter 3 for further discussion of esthetic problems.)

Occlusion, occlusal wear, and erosion

The occlusion can have significant effects on the restorative treatment plan. The following factors should be evaluated in the course of the occlusal examination: *(1)* occlusal interferences between the occlusion when the mandibular condyles are in centric relation (*centric occlusion* [CO]) and in maximal intercuspation (MI); *(2)* the number and position of occlusal contacts, as well as the stress placed on the occlusal contacts in MI; *(3)* occlusal interferences in working and nonworking excursive movements; *(4)* the amount and pattern of attrition of teeth and restorations resulting from occlusal function and parafunction; and *(5)* the interarch space available for placement of needed restorations.

Occlusal interferences. Most people have some difference between the positions of CO and MI and have no consequent pathosis. Working and nonworking contacts are also commonly noted during occlusal evaluations without consequential pathosis. These findings indicate that the existence of a discrepancy between these positions is not, in itself, an indication for occlusal equilibration. Findings from the occlusal examination that should be recorded in the restorative dentistry problem list and that do warrant treatment with occlusal adjustment are the following: *(1)* signs and symptoms of occlusal pathosis resulting from discrepancies between CO and MI (eg, mobility, excessive wear of teeth in the areas of interference between CO and MI, or periodontal ligament soreness); *(2)* occlusal discrepancies on teeth with periodontal disease; and *(3)* the need to restore the majority of the posterior occlusion.

This third factor does not imply the restoration of the majority of the posterior teeth but rather the restoration of the majority of the occlusal contacts. For example, insertion of a three-unit fixed partial denture in the mandibular right quadrant and several large restorations in the maxillary left quadrant results in the restoration of the majority of the occlusal contacts for the posterior teeth. There is no reason to fabricate the occlusion of the new restorations to duplicate the interferences that existed preoperatively. In such a case, occlusal equilibration should be completed prior to the restorative treatment. Through adjustment of only a very few occlusal contacts on teeth not involved in restorations and subsequent fabrication of the new restorations in CO, the occlusions of centric relation and MI become coincident.

Occlusal contacts. The number and position of occlusal contacts in the MI position, the force of the occlusal load, and the manner in which opposing teeth occlude in excursive function strongly influence the selection of restorative materials, as well as the design of the preparation and restoration. As the number of missing teeth

increases, so does the proportion of the occlusal load borne by each tooth. As occlusal stress increases, the dentist is forced to select the strongest of the available restorative materials and to design restorations that will provide the greatest strength in the areas of maximum stress. Likewise, the greater the potential for the patient to function on the restorations in lateral excursions, the greater the need for strength in the restorative material and the greater the imperative to select a material that will function without causing injury to the opposing dentition.

Wear. The clinician must be concerned with the abrasive potential of various restorative materials on the opposing dentition. Wear (mechanical and chemical) is a progressive phenomenon characterized by the loss of anatomical tooth form. Mechanical wear can be either *attrition* (tooth against tooth) or *abrasion* (other than tooth against tooth). Chemical wear is termed *erosion* and is the progressive loss of tooth structure through chemical processes that do not involve bacterial action. Wear may result from physiologic or pathologic causes. Physiologic wear is generally considered a slow, progressive surface degradation of tooth form manifesting as a flattening of cusp tips of posterior teeth and incisal mamelons of anterior teeth.[78,79] The mean annual physiologic occlusal wear has been estimated at 15 to 29 μm.[80,81] When wear becomes excessive, it presents restorative difficulties. Excessive occlusal wear is caused primarily by occlusal parafunction. In these instances, facets on opposing teeth match well, indicating the predominant pattern of parafunctional activity. Because altering occlusal parafunctional habits is extremely difficult, prevention of excessive occlusal wear is accomplished with the use of an occlusal appliance (Fig 2-12). The dentist should identify patients who demonstrate signs of excessive occlusal wear (especially patients who exhibit these signs at an early age) and include occlusal appliance therapy in the treatment plan.



Fig 2-12*(a)* Significant occlusal attrition caused by a habit of parafunctional grinding in a patient less than 30 years old. *(b)* An occlusal acrylic resin appliance is used to minimize the abrasive trauma generated by the parafunctional grinding habit.

The restorative materials used in dentistry today have varying abrasive potential.

No single variable is predictive of abrasivity; it is a function of a number of mechanical properties.[39] Hardness is a useful indicator, but the best predictor of wear is the relative clinical performance of the various materials. In clinical determinations of wear behavior, occlusal contact of enamel to amalgam causes only slightly greater wear to the amalgam than enamel-to-enamel contact causes to enamel. The amalgam causes less wear to the opposing dentition than does enamel.[82] The wear rate of resin composite depends on the nature of the resin composite. Microfilled composites typically exhibit similar wear behavior as enamel to abrasive forces, while micro- and nanohybrid composites may demonstrate more or less wear depending on the size of their filler particles. Micro- and nanohybrids typically generate more wear on opposing enamel than does either amalgam or enamel. Nanofilled resin composites have demonstrated mechanical properties and wear that are equivalent to many microhybrids while exhibiting improved optical properties.[83–85] Polished cast gold is more wear resistant than enamel or amalgam and generates minimal wear of opposing tooth structure.

Ceramic restorations have demonstrated a consistent ability to severely abrade the enamel of the opposing dentition[78,86] (Fig 2-13). Manufacturers of dental ceramics called *low-fusing ceramic materials* claim that they are less abrasive to the opposing natural dentition than the conventional porcelains. Several authors have supported this hypothesis[87–90]; however, just as many have contradicted it.[91–93] Some have even reported that the low-fusing porcelains can result in significantly greater enamel wear than conventional porcelain.[92,94]



Fig 2-13 Extensive tooth structure has been lost in the mandibu-lar teeth because of wear caused by the opposing porcelain fixed partial denture.

Minimizing wear of enamel by dental ceramics can best be accomplished by following these guidelines95:

- Ensure anterior guidance, which disoccludes posterior teeth in excursive movements.

- Eliminate occlusal interferences.
- Use gold alloys in functional bruxing areas.
- If occlusion is on a ceramic surface, use small-particle veneering porcelains on the occluding surfaces.
- Polish ceramic surfaces periodically.
- Adjust occlusion periodically if needed.

Occasionally, the presence of abrasive substances in the mouth is the cause of excessive occlusal wear. When the vocation or lifestyle of a patient frequently places him or her in contact with airborne abrasives, prevention of wear is difficult. Education of the patient and use of an occlusal appliance will decrease the occlusal abrasion; however, decreasing the patient's exposure to the causative agent is the only reliable means of reducing the problem.

Erosion. As defined earlier, *erosion* is the progressive loss of tooth structure through chemical processes that do not involve bacterial action. Some authors have suggested the term *corrosion* be used instead of erosion and have referred to caries as biocorrosion.[96] Erosion can result from habits such as sucking lemons or swishing carbonated beverages or from the introduction of gastric acid into the oral cavity, which can occur with repeated regurgitation. Gastroesophageal reflux disease, frequently referred to as *GERD*, occurs in the presence of an incompetent esophageal sphincter and is a common cause of acid-related erosion of the dentition. While the dentist may be the first to detect the signs of this condition, referral to a physician to manage the disease is in order. Bulimia is another condition that may be detected by the dentist first. The frequent forced regurgitation associated with this disorder results in acidic dissolution of exposed tooth surfaces and can have devastating effects on the dentition.

Chemical erosion can be distinguished from mechanical wear by the location and character of the defects. Erosive lesions have a smooth, glassy appearance, and if the process is active, the teeth are frequently hypersensitive. When found on the occlusal surfaces of posterior teeth, these lesions are characterized by concave defects into which abrasive agents are unlikely to penetrate. Severely "cupped out" cusp tips and teeth that have restorations standing above the surrounding tooth structure, often termed "amalgam islands," are clinical findings commonly associated with chemical erosion (Fig 2-14).

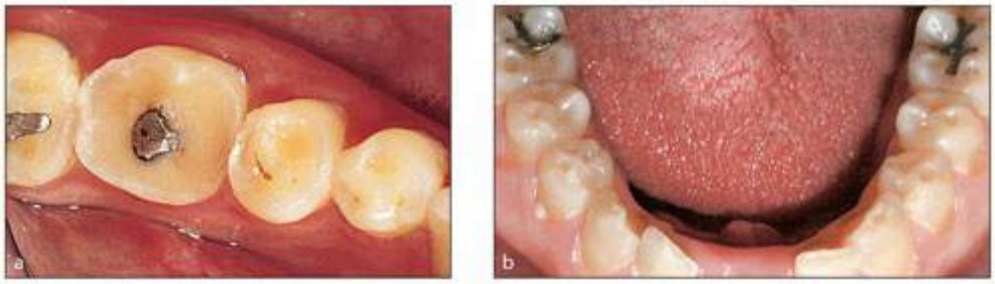


Fig 2-14*(a)* In the absence of facets that would indicate occlusal wear, significant loss of tooth structure is evidence of a chemical erosive process. Note both the amalgam restoration situated above the surrounding tooth structure and the smooth, glasslike character of the dentin. Also note the cratering pattern of the buccal cusp tips of the premolars. *(b)* In another patient, the loss of enamel on the buccal surfaces of the posterior teeth is suggestive of soft-drink swishing.

Erosion lesions appearing in both arches, primarily on the lingual surfaces of maxillary teeth and the occlusal surfaces of posterior teeth, are characteristic of erosion caused by gastric acid. Smooth lesions on the facial surfaces might be of chemical or mechanical origin. In instances of uncertainty, questions related to habits may elucidate the cause of mechanical abrasion, while a thorough history and medical evaluation may reveal the presence of acid-related erosion. When bulimia is the underlying problem, detection is often difficult. The dentist must be tactfully candid in discussing this possible etiology. The primary cause of the loss of tooth structure should always be determined and resolved before rehabilitative therapy is undertaken.

Often wear lesions are the result of a combination of attrition, abrasion, and erosion. Once enough enamel has been lost to expose the dentin, erosion and *three-body wear*, a form of abrasion, can severely exacerbate the loss of tooth structure. This happens commonly on the cusp tips and incisal edges, resulting in deep cratering or fissuring and a potential for enamel chipping and fracturing. Resin composite offers the opportunity to successfully restore these noncarious lesions with little or no preparation with a bur.[97] Early intervention in these lesions can prove a valuable service to patients (Fig 2-15). A thorough review of the loss of tooth structure from wear and erosion has been published by Verrett.[79]



Fig 2-15*(a)* Cratered noncarious occlusal lesions on the buccal cusps of both mandibular premolars. *(b)* Restoration of these lesions with a microhybrid resin composite.

Interarch space. When the dentist determines that significant loss of occlusal tooth structure has occurred and pulpal sensitivity has arisen, or that teeth have been so weakened by abrasion or erosion as to be at risk for fracture, restorative treatment is indicated. The dentist must evaluate the occlusion in MI and determine whether sufficient space exists for the placement of the restoration. If inadequate space is available, the dentist must either *(1)* gain space for adequate tooth reduction and restoration resistance form by surgical crown lengthening, orthodontic intrusion, or shortening the opposing tooth or *(2)* select a different restorative option that requires less bulk of material for resistance. Recognition of the space inadequacy prior to tooth preparation is essential.

In those cases in which generalized wear or erosion has resulted in the loss of an extensive amount of tooth structure, the dentist is faced with a significant restorative problem. In these instances, sufficient interarch space is often not available to restore the lost tooth structure without increasing the vertical dimension of occlusion. This represents a complex restorative process involving more than a consideration of the mechanics of individual tooth restoration.

Axial tooth surfaces

Unlike changes in occlusal contours, the alteration of the axial contours of teeth is not caused by tooth-to-tooth wear.

Although it is generally due to erosion or abrasion, occlusally generated stresses may contribute to this phenomenon in some instances.[98] The term *abfraction* is applied when noncarious cervical lesions (NCCLs) are thought to have a combined cause of abrasion and occlusally induced tooth flexure.[99–102] There is general

agreement that the etiology of NCCLs is multifactorial,[103] Preventive treatment for cervical abrasion is directed at altering the habit or other factor(s) causing the problem. Modification of tooth brushing habits to include the use of small, pea-sized amounts of minimally abrasive toothpastes with a neutral pH can reduce the rate of erosion and abrasion. If abfraction is suspected, treatment should include the nighttime wear of an occlusal appliance.

NCCLs should be included on the problem list to alert the patient to the problem and to ensure that the dentist addresses the possible causes and considers restorative treatment options. In the absence of symptoms, the extent of the lesion should be assessed and restorative intervention should be a matter of clinical judgment. A prudent approach would be to restore the area when tooth loss has progressed to the point that the normal tooth contour could be replaced with restorative material without leaving the restorative material too thin to withstand functional and abrasive stresses. The reader is referred to a 2003 paper on diagnosis and treatment of NCCLs[104] and to chapter 15 for a detailed discussion of etiology and treatment.

Tooth integrity and fractures

Tooth fractures are either complete or incomplete. A 2004 study[105] presented the risk indicators and incidence of complete cusp fractures in posterior teeth. The authors concluded that fewer than 10% of complete cusp fractures of posterior teeth occur in teeth without restorations and that the greatest risk indicator is the presence of a fracture line that is detectable through tactile examination. Lubisich et al[106] reported on the frequency and location of cracked teeth by averaging the results of 12 studies. Mandibular molars comprised 48% of the cracked teeth, while maxillary molars made up 28%. Maxillary premolars were 16% of the cracked teeth, and mandibular premolars were found to have cracks much less frequently at 6%. In the maxillary arch, the first molar and first premolar fractured slightly more frequently than the second molar and premolar. In the mandibular arch, the first molar cracked about twice as often as the second molar. In both arches, the nonholding cusps (maxillary facial cusps and mandibular lingual cusps) tended to fracture more often than the holding cusps. This trait was more pronounced in the mandibular arch.[107,108]

Incomplete tooth fractures are most commonly called *cracked teeth*, but several terms have been used over the years.[109,110] Cracked-tooth syndrome is a fairly common result of the incomplete fracture of a vital tooth. Patients suffering from cracked-tooth syndrome present with a series of symptoms that include discomfort during chewing, unexplained sensitivity to cold, and pain on application or release

of pressure.[111–117] Cracked-tooth syndrome may be found in restored or unrestored teeth.[118] In restored teeth, it is often associated with existing small to medium-sized restorations.[119,120] A practice-based study of 1,962 molars found that molars with resin composite restorations had 4 times the likelihood of having a crack compared with an unrestored molar, while a molar with an amalgam restoration had 7.7 times the likelihood of having a crack compared with an unrestored molar.[121] Another study of 51 patients concluded that teeth treated with Class 1 or 2 restorations have 29 times greater risk for cracks.[116] Often patients with multiple cracked teeth have parafunctional habits or malocclusions that have contributed to the problem. Cracked-tooth syndrome is an age-related phenomenon; the greatest occurrence is found among patients between 33 and 50 years of age.[118] A recent study of the etiology for increased incidence of incomplete and complete fractures includes intraoral jewelry, especially in the tongue.[122] Lip piercings have been associated with an increase in gingival recession and a slight increase in localized periodontitis.[123]

Cracked-tooth syndrome is often difficult to diagnose. The patient is frequently unable to identify the offending tooth, and evaluation tools, such as radiographs, visual examination, percussion, and pulp tests, are typically nondiagnostic. The two most useful tests are transillumination and the biting test.

Many teeth contain cracks and craze lines, most of which cause no symptoms; however, transillumination of a severely cracked tooth generally presents a distinctive appearance that permits the clinician to distinguish minor cracks from those deep enough to result in symptoms. When a tooth with a severe crack (one that extends into dentin) is transilluminated from either the facial or lingual direction, light transmission is interrupted at the point of the crack.[106] This results in the portion of the tooth on the side away from the light appearing quite dark.

The transition from bright illumination on one side of the tooth to darkness on the other is sudden rather than gradual, occurring abruptly at the point of the fracture (Fig 2-16). The biting test is the most definitive means of localizing the crack responsible for the patient's pain. By having the patient bite a wooden stick, rubber wheel, or one of the commercially available instruments designed for that purpose (eg, Tooth Slooth, Professional Results), the dentist is generally able to reproduce the patient's symptom and identify not only the cracked tooth but also the specific portion of the tooth that is cracked. Crunchy food placed sequentially on suspected teeth has also been suggested as a diagnostic aid.[124] Once the offending tooth has been identified, tooth preparation often allows visualization of the crack (Fig 2-17;

see also Fig 1-8).



Fig 2-16 With the use of transillumination, cracks in the tooth are visible through the mesial and midlingual areas when the transmission of the light is disrupted.

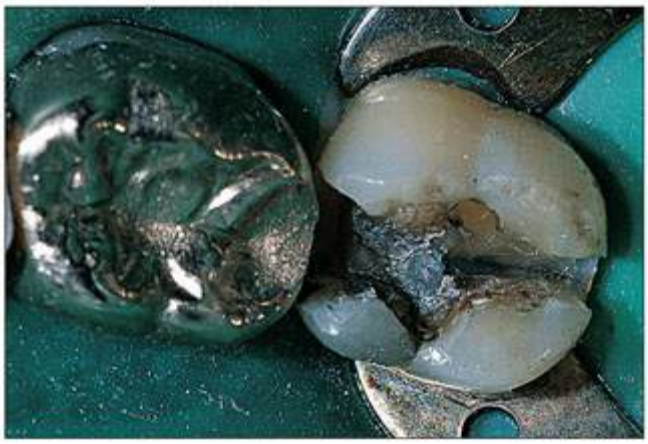


Fig 2-17 The mesiodistal crack in the pulpal floor of the mandibular right second molar caused sharp pain upon chewing. The tooth is to be restored with an onlay to splint the tooth together during function, relieve the patient's symptoms, and prevent propagation of the crack.

Where direct diagnostic methods prove unsuccessful, indirect methods may be used. Orthodontic bands may be placed on suspected teeth to prevent separation of the crack during function. If the patient's symptoms subside, the diagnosis of cracked-tooth syndrome can be made.

In the treatment of incomplete tooth fracture, the tooth sections are splinted together with a complete-cuspal-coverage restoration.[114,115,119,125] Although a full-veneer crown is often the treatment of choice, cuspal coverage and protection may also be accomplished with the use of an amalgam restoration[126] or an indirectly fabricated onlay of metal, ceramic, or resin composite. Because of their potential to lose bond integrity over time, bonded intracoronal restorations presently have not been considered to be adequate for long-term resolution of the problem.[115,127,128]

However, several recent studies have demonstrated success with direct resin composite intracoronal or cuspal overlay restorations.[129–131] After their thorough review of the literature, Lubisich et al[106] concluded that cracked teeth with reversible pulpitis could be successfully treated with a full-coverage crown, complex amalgam, or a bonded ceramic or resin composite onlay.

While the diagnosis of incomplete tooth fracture has historically been symptom based, the dental operating microscope, with its high magnification capability, allows the dentist a new level of increased diagnostic sensitivity. An excellent review on the use of the surgical optical microscope in the diagnosis of early enamel and dentin cracks is available.[132]

Esthetic evaluation

In addition to an esthetic evaluation of existing restorations, an assessment of the esthetics of the entire dentition should be completed. Because dental esthetics is a subjective area, patients should be questioned about any dissatisfaction they may have regarding the esthetics of their dentition. In the absence of complaints by the patient, the dentist's impressions regarding esthetic problems should be tactfully conveyed to determine whether the patient would like the esthetic problems addressed. The dentist is often better able than the patient to determine how dental procedures might enhance the patient's appearance. If an agreement is reached between the patient and dentist as to the existence of specific esthetic problems, the problems should be included on the restorative dentistry problem list.

Commonly encountered esthetic problems that are related to or may be addressed by restorative dentistry include *(1)* stained or discolored anterior teeth; *(2)* unesthetic contours in anterior teeth (eg, unesthetic length, width, incisal edge shape, or axial contours); *(3)* unesthetic position or spacing of anterior teeth; *(4)* caries lesions and unesthetic restorations; *(5)* excessive areas of dark space in the buccal corridors due to a constricted arch form; and *(6)* unesthetic color and/or contour of tissue adjacent to anterior restorations. This last problem includes excessive gingival display, occasionally referred to as the "gummy smile." (See chapter 3 for a thorough discussion of esthetic considerations in diagnosis and treatment planning.)

The restorative treatment of esthetic problems may range from conservative therapy, such as bleaching, to more invasive measures, such as the placement of resin composite or ceramic veneers and posterior restorations or complete-coverage crowns. Additionally, adjunctive periodontal, endodontic, or orthodontic procedures may be helpful, depending on the nature of the original problem. Esthetic

restorations are discussed in subsequent chapters.

Evaluation of the periodontium

From a restorative dentistry perspective, the periodontium must be evaluated primarily for two reasons: *(1)* to determine the effect that the periodontal health of the teeth will have on the restorative dentistry treatment plan and *(2)* to determine the effect that planned and existing restorations will have on the health of the periodontium.

Evaluation of the periodontium consists of a clinical assessment of attachment levels, bony topography, and tooth mobility; a qualitative assessment of tissue health; and a radiographic evaluation of the supporting bone. The assessment of attachment levels involves periodontal probing of the entire dentition with both a straight probe for determination of vertical probing depths and a curved probe to explore root concavities and furcation areas. Any bleeding induced by gentle probing should be noted. A variety of tests are available to aid in determining the presence and identity of periodontal pathogens; however, the most consistent clinical indicator of inflammation is bleeding on probing.[133] Bleeding on probing does not always indicate the presence of active periodontal disease, but active disease has been consistently found to be absent when there is no bleeding on probing.[133]

The qualitative assessment of periodontal tissue health calls for a subjective assessment of the inflammatory status of the tissue; tissue color, texture, contours, edema, and sulcular exudates are noted. The presence of specific local factors, such as plaque and calculus, and their relationship to tissue inflammation should be noted. Abnormal mucogingival architecture, such as gingival dehiscences and areas of minimal attached gingiva, should be recorded. This is especially true when these anomalies are noted in the proximity of existing or planned restorations.

During examination of the periodontium, the dentist not only must be cognizant of periodontal inflammation adjacent to existing restorations but also must estimate the location of margins for future restorations and their potential for impinging on the biologic width. Review of radiographs, especially correctly angulated bitewing radiographs, during the periodontal examination enables the dentist to assess the relationship of existing and planned restorations to bone levels and to correlate radiographic signs with clinical findings.

When the clinical and radiographic portions of the periodontal examination have been completed, a periodontal prognosis should be established for all teeth; special attention should be given to teeth involved in the restorative dentistry treatment plan.

Teeth requiring restorative treatment that have a guarded periodontal prognosis should be noted in the restorative dentistry problem list. Until the periodontal prognosis becomes predictably positive, the restorative treatment of teeth with a guarded prognosis should be as minimal as possible, and treatment planning that relies on these teeth must remain flexible.

Evaluation of radiographs

The radiographic examination is an essential component of the comprehensive evaluation. Problems detected during the evaluation of radiographs should be listed under the appropriate headings on the problem list (eg, restorative, endodontic, periodontal).

Although radiographs can provide valuable information for use in diagnosis and treatment planning, exposure of patients to ionizing radiation must be minimized; therefore, discretion is required when the dentist orders radiographs. There are no inflexible rules for radiographic evaluation; rather, clinical judgment should be exercised. The goal is to minimize unnecessary exposure and cost but to avoid underutilization, which could result in inadequate diagnosis. The use of patient-specific criteria is the key. Different patients have different requirements both in terms of the radiographic views needed and the frequency with which radiographs should be repeated.

A reasonable guideline to follow is that all dentate patients should initially have a radiographic series completed that reveals the periapical areas of the entire dentition. This will permit detection of central lesions not visible on bitewing radiographs and will serve as a baseline, allowing the clinician to assess changes over time. Bhaskar[134] has reported that approximately 85% of central jaw lesions are apparent in views of the apical areas of the dentition but are not visible on bitewing radiographs. For patients with periodontal disease, periapical and/or vertical bitewing radiographs are indicated. For patients with no significant periodontal pathoses, a panoramic radiograph provides the necessary view. For all patients with approximating teeth, a series of films is indicated to show the proximal areas of posterior teeth. Bitewing radiographs serve this purpose.

The frequency with which radiographs should be updated is a matter of clinical judgment. The dentist should assess the etiologic factors present and determine whether new disease is likely to have occurred since the last radiographic examination. The dentist must weigh the risk of undetected disease against the cumulative risk of radiation exposure. A suggested guideline is to make new

bitewing radiographs of caries-active adults at 6- to 18-month intervals and of caries-inactive adults every 2 to 3 years.[135,136] Patients may be considered minimally susceptible to caries if they have had no caries lesions in recent years, demonstrate low plaque levels, have adequate salivary flow, have a noncariogenic diet, and exhibit no clinically discernible caries lesions. Periapical radiographs of the entire dentition should be repeated only as dictated by the specific needs of the treatment to be accomplished. For example, a patient under active treatment or maintenance for periodontal disease may require an updated radiographic series every 2 to 3 years to reevaluate bony contours, while another patient whose disease processes are controlled may require subsequent periapical radiographic updates only every 4 to 5 years.

In evaluating radiographic findings for restorative purposes, the dentist should note open interproximal contacts, marginal openings, overhanging restorations, periapical radiolucencies and radiopacities, and radiolucencies within the body of the tooth or the restoration (see Figs 2-4 and 2-6). The dentist must interpret "abnormal" radiographic findings with caution. Many phenomena that are detectable radiographically can also be detected clinically and should be verified clinically before treatment is planned. This is especially true when the clinician evaluates radiolucencies that appear to represent carious tooth structure but may in fact represent nonpathologic structures. An example of this is the radiographic phenomenon commonly known as "burnout" (Fig 2-18). *Burnout* is a radiolucency that is not caused by demineralization but instead occurs when the x-ray beam traverses a portion of the tooth with less thickness than the surrounding areas. It is most commonly found near the cervical area of a tooth and may be caused by concavities in the tooth or the angulation of the beam, but it is also related to the portion of the tooth not covered by enamel or alveolar bone.

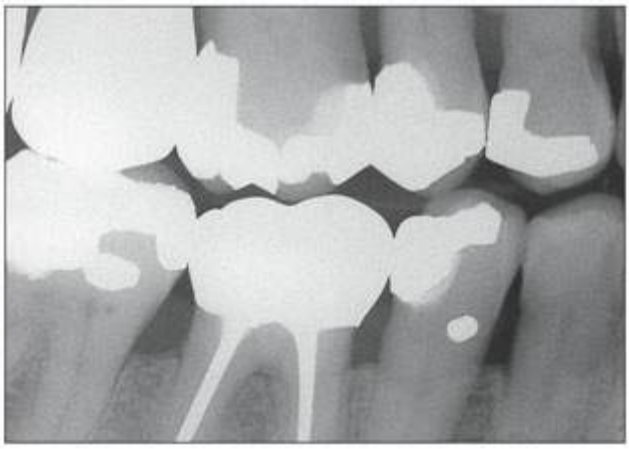


Fig 2-18 The radiolucent area beneath the restoration in the mesial surface of the maxillary first molar is radiographic burnout. No carious tooth structure is present.

The dentist must be careful not to mistakenly diagnose as demineralized tooth structure a decrease in radiopacity resulting from an abraded area. Likewise, the dentist must be cautious in diagnosing carious tooth structures beneath existing restorations, because certain radiolucent dental materials have a radiographic appearance similar to that of demineralized tooth structure. A comprehensive review of dental radiology has been provided by Goaz and White.[137]

Evaluation of diagnostic casts and photographs

The dentist can gain valuable information through an evaluation of diagnostic casts and patient photographs. The use of diagnostic casts and photographs allows the dentist to thoroughly analyze conditions while the patient is not in the chair. By examining diagnostic casts of the dentition, the dentist can see areas that are visually inaccessible during the clinical examination. Facets and marginal openings that may be difficult to discern intraorally are readily visible on the diagnostic casts. Facets on the casts of the dentition can be aligned to provide a guide to dynamic occlusal relationships. In addition, the dentist may use gypsum casts to complete diagnostic preparations and diagnostic wax-ups, simulating planned treatment. Where removable partial dentures are indicated, survey and design procedures may be completed on the diagnostic casts before restorative treatment is planned. The requirements of removable partial denture design may thus be considered during the planning of restorative care.

Although not every case requires the evaluation of casts mounted on a semiadjustable articulator, cases involving significant occlusal wear, multiple missing teeth, or the restoration of a significant portion of the occlusion should be evaluated with diagnostic casts mounted in centric relation. If multiple teeth are missing, the articulator maintains the correct interarch relationship, permitting buccal and lingual views of interarch spaces. Using a semiadjustable articulator that provides a reasonable approximation of the patient's intercondylar distance, condylar inclination, lateral guidance, and hinge axis of rotation, the dentist can simulate the patient's mandibular movements. This enables the clinician to assess the occlusal scheme and to plan restorative care accordingly.

Treatment Plan

Having completed a comprehensive examination, the dentist documents problems related to restorative dentistry on the restorative dentistry problem list (Fig 2-19).

Each problem on the list is then reevaluated. After consideration, some of the problems may be deleted. For example, a tooth with a defective restoration may also have a significant loss of periodontal attachment and, therefore, a poor periodontal prognosis. In such a case, the defective restoration is initially considered a problem, but, in view of the periodontal condition, the tooth would be planned for extraction rather than restoration. The defective restoration is then omitted from the restorative problem list. Sometimes the treatment planned to address a problem may lead to additional problems. For example, reducing an extruded tooth to the level of the occlusal plane may result in the need for elective endodontic treatment, surgical crown lengthening, and a full-coverage restoration.

PATIENT: Blank, Felina D.

PROBLEM LIST

Chief complaint: "My tooth hurts every time I chew, and lately iced tea has made it hurt, too."

Medical/systemic: Hypertension. Present blood pressure: 155/95

Restorative (also see charting):

- Incomplete tooth fracture of mesiolingual cusp, #19
- Caries lesions, #20, #21, #28 (high caries risk)
- Defective restorations, #2, #12
- Facial, noncarious cervical lesion, #12
- Worn incisal edges, #6 to #11
- Fluorosis stain, #8
- Biologic width impingement, #3, distal
- Patient wishes to whiten maxillary anterior teeth

Periodontal:

- AAP Case Type I (see periodontal charting form)
- Generalized marginal gingivitis
- Generalized minimal bone loss with 3- to 4-mm pockets
- Vertical defect, #3, distolingual (5 mm)
- Biologic width problem, #3, distal
- Plaque and calculus: Generalized interproximal plaque in all posterior sextants (modified O'Leary index: 50% plaque free), subgingival calculus revealed on bitewing radiographs of #19 and #30; supragingival calculus present on lingual surfaces of mandibular anterior teeth

Endodontic: None

Prosthodontic: Missing, #29

Orthodontic: None

Occlusion: Supraeruption, #4; excessive wear, #6 to #11

Temporomandibular dysfunction: None

Oral surgery: None

Fig 2-19 Example of a problem list.

Once the final problem list is formulated, the next step is to establish a plan for the treatment of each problem on the list. The treatment planned for each problem should be based on current research evidence to the extent possible.[138] Caries should be treated as a disease using a medical model,[139,140] and interventions to stop demineralization and bring about remineralization should be planned for early caries lesions.[140–142] (See chapter 5 for an in-depth review of current strategies for caries management.) A problem list worksheet (Fig 2-20) is a useful tool to help organize

the planning of treatment for each problem. It consists of an unsequenced list of problems and their associated solutions. Later, during the sequencing process, this list of treatments will be integrated into the comprehensive treatment plan.

PATIENT: Blank, Felina D.

PROBLEM LIST WORKSHEET

Problem	Treatment
Chief complaint: cracked #19	• Gold onlay
Hypertension	• Referral to physician for evaluation and treatment
Caries	• Educate patient: snacking, hygiene techniques, home fluoride use • Rx: neutral sodium fluoride (1.1%) gel or dentifrice • If caries continues, complete caries risk assessment (diet survey, *S mutans* culture) • #20, #21: Class 5 resin-modified glass-ionomer restorations
Defective restorations	• #2: Mesio-occlusodistal (MOD) amalgam • #4: Porcelain-fused-to-metal (PFM) crown (shorten to level occlusal plane) • #12: Mesio-occlusal resin composite
Abrasion: #12	• Class 5 resin composite restoration
Wear: #6 to #11	• Protective acrylic resin occlusal splint
Fluorosis: #8	• Microabrasion
Biologic width: #3	• Surgical crown lengthening
Patient desires to lighten maxillary anterior teeth	• Home bleaching, #5 to #12
Periodontal inflammation associated with local factors	• Patient education and hygiene instruction; goal: 90% plaque-free index • Prophylaxis; scaling/root planing in mandibular sextants and any areas not responding to initial care • Reevaluate; goal: eliminate bleeding on probing • Surgical crown lengthening, #3: osseous recontouring and soft tissue excision
Missing: #29	• Fixed partial denture (FPD), #28 to #30; PFM retainer, #28; three-quarter retainer, #30
Supraeruption: #4	• Shorten #4 when PFM crown is completed

Fig 2-20 Example of a problem list worksheet to accompany the problem list shown in Fig 2-19.

Planning the restoration of individual teeth is the "nuts and bolts" of restorative dentistry treatment planning. It requires the consideration of four primary factors as well as a number of modifying factors. The primary considerations are *(1)* the amount and form of the remaining tooth structure, *(2)* the functional needs of each

tooth, *(3)* the esthetic needs of each tooth, and *(4)* the final objective of the overall treatment plan.

Remaining tooth structure

The quantity and location of remaining tooth structure determine the resistance features available for the restoration and thus greatly influence the restorative design. These factors determine not only the resistance to displacement of the restoration but also the fracture resistance of the remaining tooth structure. The clinician should select the restoration that provides for the best retention and the optimal protection of the remaining tooth structure, using the least invasive design possible.

For the restoration of posterior teeth, an intracoronal restoration with amalgam or resin composite is generally the most conservative choice, and both materials have proven to be clinically successful. When the width of the intracoronal preparation of a posterior tooth exceeds one-third of the intercuspal width, the tooth becomes significantly more susceptible to cuspal fracture and the concern becomes not only restoration failure but also tooth fracture.[143]

Even more significant than restoration width to the susceptibility of a tooth to fracture is the depth of the preparation.[144] In instances of deep and/or wide preparations, the clinician must assess the need for occlusal coverage to protect the fracture-prone portions of the tooth. Often the assessment of remaining tooth structure and integrity can be accomplished only after removing the existing restoration and any defective tooth structure. For occlusal coverage, choices include cuspal-coverage amalgam, partial veneer restorations (eg, onlays, three-quarter crowns, or seven-eighths crowns), and complete crowns. The clinician should resist the temptation to progress immediately to a complete crown and instead should select the most conservative choice that satisfies the needs of the individual tooth and the overall treatment plan.

The quantity of remaining tooth structure has an equally important effect on the choice of restorations for anterior teeth. For conservative interproximal restorations in anterior teeth, resin composite is almost always indicated because sufficient tooth structure is generally available for effective resin-enamel and resin-dentin bonding. When extensive facial tooth structure has been esthetically compromised but the facial enamel and the majority of the lingual aspect of the tooth remain intact, a ceramic veneer affords a conservative alternative to a complete crown. The veneer satisfies the esthetic requirement but is considerably less invasive than complete

coronal coverage. When the facial enamel has been destroyed, significant lingual tooth structure has been lost, occlusal stress is exceptionally heavy, or there is very darkly pigmented dentin, veneers are not a viable option and complete crowns are required (Fig 2-21).



Fig 2-21 Facets and chipped incisal edges are evidence of the severe stresses placed on these anterior teeth by parafunction. Although they demonstrate tetracycline staining and possess a largely intact facial surface, these teeth would be poor candidates for veneer restorations. Complete-coverage restorations are indicated if the patient wishes to mask the tetracycline staining.

Functional needs

The choice of restorative materials and the design of restorations must accommodate the functional needs of the individual patient. This precludes the use of a cookbook approach to treatment planning and requires that the clinician assess the specific circumstances of each tooth before planning restorative procedures. The functional and parafunctional stresses of the occlusion are significant considerations in this decision process. For example, a patient with average-strength musculature, an anterior-guided disocclusion of posterior teeth in excursions, and minimal tendency toward parafunction may require only an intracoronal amalgam or resin composite restoration to restore mesial and distal surfaces of a posterior tooth. In a similar circumstance, a patient with heavy musculature, signs of parafunctional activity, and no anterior-guided disocclusion may require a cast metal restoration that covers the entire occlusal surface to minimize the chances of tooth fracture.

A useful guide in making decisions about material selection and restoration design is the evidence of functional demand provided by the existing dentition. Patients who present with a dentition exhibiting minimal destruction are good candidates for conservative, directly placed restorations. Patients whose teeth exhibit severe wear facets or considerable loss of tooth structure from occlusal attrition are best served by materials high in strength and wear resistance, such as cast metal or possibly

zirconia restorations.

Restorations planned for NCCLs pose little dilemma in terms of restorative choices. Any of the restorative materials suited to the restoration of Class 5 areas will serve satisfactorily. Glass-ionomer restorative materials have proven to be effective in the restoration of Class 5 areas, providing longevity in excess of 10 years.[145,146] Resin-modified glass-ionomer restorative materials provide an alternative to conventional glass-ionomer cements and have been shown to demonstrate exceptional retention. One study[147] found a retention rate of 98% after 3 years, and an independent study[148] found 100% retention after 5 years. Improvements in the performance of resin adhesives have made the retention of resin composites and polyacid-modified resin composite restorations predictable as well. A number of investigators have reported retention rates for resin composite restorations of over 95% after 3 years.[149,150] Another study reported a resin composite retention rate of 98% after 18 months, but over the following 2 to 3 years the loss rates increased considerably, resulting in a 75% retention rate after 8 years, with no significant difference in retention whether bonding to nonsclerotic or sclerotic dentin.[151] A comparative study of six resin bonding systems and a resin-modified glass ionomer with annual evaluation over a 13-year period concluded that all systems showed a continuous degradation of the bond, with variations dependent on the adhesion strategy. The best retention (about 65%) was achieved with the resin-modified glass ionomer, followed closely by a four-step etch-and-rinse system (about 64%).[146] The glass-ionomer materials offer the anticariogenic advantage of fluoride release, while the materials containing greater amounts of resin composite generally provide better esthetics and wear resistance.

Another strategy for the treatment of NCCLs involves a connective tissue graft or a coronally advanced flap alone or in combination with a resin-modified glass-ionomer restoration. Six-month to 2-year root-coverage results of better than 80%, and some as high as 90%, have been demonstrated. The amount of gingival coverage was unaffected by the presence of a resin-modified glass-ionomer restoration, demonstrating the compatibility of this material with the periodontium.[152,153]

The patient's level of caries activity will influence the selection of restorative materials. Patients whose caries risk assessment indicates a high potential for caries are good candidates for treatment with anticariogenic restorative materials as well as the use of a caries-management protocol.[154] Conventional glass-ionomer cements have been found through clinical study to provide an anticariogenic effect.[54–58] Resin-modified glass-ionomer materials have been found to inhibit simulated caries

in vitro[47–51] and have been shown to possess anticariogenic properties in the clinical environment.[52,55,56,155] None of the anticariogenic restorative materials presently available are able to withstand the stresses of occlusal function. In addition, in cases of severe caries challenge, such as occurs with patients having greatly diminished salivary flow due to medications or radiation therapy, exposure to additional sources of fluoride beyond that which can be provided by these dental materials is needed to maintain tooth health.[56,157]

Esthetic needs

Establishing the patient's esthetic priorities is essential in planning restorative care. In most instances, the dentist will have the choice of a tooth-colored or a non–tooth-colored restoration for a given situation. Because metals are generally superior in strength and durability, they are the materials of choice when strength and wear resistance are the overriding considerations. With the patient's input, the clinician must decide which requirement is more important, durability or esthetics.

For intracoronal, directly placed restorations in the anterior area of the mouth, resin composites are the obvious choice. They can be made to match most teeth in color and have been shown to be acceptable with a success rate of up to 95% after 5 years; the reasons for failure are surface discoloration, caries, and fracture of the restoration.[157–160] In stress-bearing areas in the posterior aspect of the mouth, amalgam has been the material of choice for direct intracoronal restorations.[161,162] When compared with large amalgam restorations, large resin composite restorations have not always fared as well in clinical studies as do the more conservative restorations.[163,164] Where full or partial occlusal coverage is required, amalgam has been found to yield favorable results, routinely providing service in excess of 10 years[165–167] (Fig 2-22). Recent studies, however, have demonstrated that large, multisurface resin composite restorations are performing as well as multisurface amalgam restorations in excess of 10 years of service.[168–170] In the high-caries-risk patient, amalgam remains the restoration of choice and can be a predictable alternative to crowns.[167,168]



Fig 2-22 Example of a large cuspal-coverage amalgam restoration. This restoration is 6 years old.

Cast metal restorations offer even greater longevity and should receive special consideration for patients with parafunctional habits (see chapter 20).[171–173] Although these indirect restorations may have accompanying complications, they offer an excellent service to the patient in maintaining tooth function.[174] When occlusal coverage is required in an area of esthetic concern, the clinician may choose between an all-ceramic and a metal-ceramic restoration. All-ceramic restorations generally provide a superior esthetic result, while the metal substructure of metal-ceramic restorations offers tremendous strength. However, recent improvements in the fabrication techniques of all-ceramic restorations have produced high-purity copings that have the potential to replace the metal copings for the metal-ceramic crowns. These copings are made through a slip-casting technique (In-Ceram Alumina, Vident) or a computer-aided milling process (Procera AllCeram, Nobel Biocare).[175] These crowns have demonstrated flexural strengths four to six times greater than conventional feldspathic and pressed porcelain systems.[176] The strongest of the ceramic cores is zirconia, which has a strength that approaches seven times that of conventional porcelains.[39]

Unfortunately, as the strength of the core ceramic increases, the esthetic value decreases, but this may be more important for teeth in the esthetic zone than for posterior teeth. A moderate-strength ceramic core, lithium disilicate (eMax [formerly IPS Empress 2], Ivoclar Vivadent), has good esthetics with or without veneering porcelains and is also four times as strong as traditional porcelains and about one-half as strong as zirconia.[39,176] Giordano[176] reports a success rate of about 97% for Procera from 2-year data in the United States and 5-year data from Europe and a 98% success rate for In-Ceram from 7-year data with anterior and posterior crowns. Other studies evaluating crown survival and clinical success provide favorable data for these stronger ceramic systems.[177–180]

Zirconia-based restorations are rapidly becoming the indirect restoration of

choice by many practitioners. They not only offer esthetics, biocompatibility, color stability, wear resistance, relatively low cost, and low thermal conductivity but also possess comparable strength to porcelain-fused-to-metal (PFM) restorations. Zirconia-based crowns and fixed partial dentures (FPDs) can be conventionally cemented in a manner similar to that for PFMs. Advances in computer-aided design/computer-assisted manufacture (CAD/CAM) technology have made the use of zirconia more readily available in dentistry. Zirconia has been recommended as suitable for posterior or anterior crowns, long-span FPDs, and implant abutments.[181] The primary concern with zirconia restorations has been the chipping and fracturing of the veneering porcelains. However, more refined processing procedures have started to yield much better results in preventing the loss of the surface porcelain.[182,183]

All-ceramic materials and fiber-reinforced resin materials have been marketed for use in the fabrication of FPDs. One clinical study reported a 93% success rate for 30 posterior lithium disilicate–based core ceramic FPDs after 2 years.[184] In a retrospective study of In-Ceram Alumina–core FPDs, Olsson et al[185] found successful results for anterior FPDs and promising survival rates (93%) for alumina-core posterior FPDs during a 5-year observaion period. Another study demonstrated a 90% success rate after 5 years with In-Ceram Alumina posterior FPDs.[186] Another promising technology is the computer-aided milling of single, large zirconia blanks for the cores of long-span posterior prostheses. A recent in vitro study reported on posterior zirconia-based FPDs spanning up to five units with a life expectancy comparable to metal-ceramic FPDs.[187] Although these results appear promising, more long-term clinical data is required to support the use of all-ceramic core materials as an alternative to metal or metal-ceramic materials for posterior FPDs. With regard to the single missing tooth, osseointegrated dental implants are replacing the traditional FPD as the treatment of choice in appropriate clinical situations.

Ultimately, predictably successful restorations depend on the choice of a system that fits the clinical situation, proper preparation of the tooth, careful laboratory fabrication of the restoration, and careful insertion technique.[188,189] Even though there has been marked improvement in the strength of all-ceramic restorations, they are still subject to fracture if proper preparation principles are not followed (Fig 2-23) . Chapters 17 to 20 are dedicated to the discussion of advantages and disadvantages of the different designs and materials of indirect restorations.



Fig 2-23 Example of a catastrophic failure of an all-ceramic crown. The crown is 5 years old. The failure is probably due to inadequate tooth preparation of the palatal cusp and excessive occlusal forces. The thin porcelain is visible at the mesial aspect of the fracture.

Final treatment objective

The anticipated ultimate outcome of restorative and prosthodontic rehabilitation is the final factor to consider when the design of a restoration is planned and the restorative material is selected. Teeth that may require one type of restoration to restore health and function may require a different treatment to meet the needs of the final treatment plan. For example, if implant treatment is planned or if no prosthodontic replacement is planned for teeth that are missing, the teeth adjacent to the edentulous area may require only conservative restorative care for the treatment of small caries lesions. In a different treatment plan that calls for replacement of the missing teeth with a removable partial denture, surveyed indirect restorations may be required for the teeth adjacent to the edentulous area. In a third variation of the same case, missing teeth may be replaced with an FPD and the teeth in question may be needed as FPD abutments.

When the final treatment objective has been visualized, it is often possible to identify certain teeth as key teeth, the retention and restoration of which are crucial to the success of the treatment plan. These teeth are often potential prosthodontic abutments and/or canine teeth, which are typically critical for developing an appropriate occlusal scheme in the final restorative rehabilitation. Because the success of the total treatment plan often hinges on these teeth, it is crucial to ascertain their periodontal and endodontic prognosis and to plan the restorative treatment that provides the best long-term prognosis. This may dictate an aggressive restorative design to achieve the most predictable success for these key teeth.

The following example serves to illustrate this principle. A hypothetical patient has a freestanding second molar that contains a defective mesio-occlusodistal (MOD) amalgam restoration. Although the facial wall location slightly undermines

the facial cusp, a replacement amalgam restoration appears likely to function adequately. In the comprehensive treatment plan, the tooth will serve as a distal abutment for a removable partial denture. With mere replacement of the defective amalgam restoration, the tooth is at some risk for cuspal fracture in the future. Fracture of the tooth would necessitate fabrication of a crown beneath the removable partial denture. By planning a crown prior to fabrication of the removable partial denture, a treatment plan somewhat more aggressive than would be dictated by the needs of the individual tooth, the prognosis for the ultimate treatment objective becomes more predictable and the risk of compromising the final result is reduced. This does not mean that every removable partial denture abutment should receive a crown, but it is intended to convey the importance of planning for predictable longevity in key teeth.

Treatment Sequence

Once the completed treatment has been envisioned and the design of the restorations required to address each problem on the restorative dentistry problem list has been established, the final step in establishing the restorative treatment plan is sequencing the treatment, which can present a significant challenge. Proper sequencing of all procedures involved is critical to successfully achieving the treatment goals. The dentist not only must be able to envision the ultimate outcome of treatment but must also understand the order in which the proce- dures must be performed to achieve that outcome. Considerable thought is required to understand and plan a treatment sequence that avoids unnecessary complications.

Most restorative treatment will fall into the categories of disease control or definitive rehabilitation treatment. Restorative treatment aimed at controlling active disease generally consists of direct restorative procedures using amalgam, resin composite, or glass-ionomer materials. Assuming that the patient's chief complaint has been satisfactorily addressed, the sequence of treatment within the disease control phase is dictated by three considerations: *(1)* severity of the disease process (ie, the most symptomatic tooth, the tooth with the deepest lesion, or the most debilitated tooth is restored first), *(2)* esthetic needs, and *(3)* effective use of time. At each appointment, treatment is rendered in the area in the most acute need of treatment. When possible, the restorations should be completed by sextant or quadrant to optimize the use of time.

Treatment provided in the definitive rehabilitation phase goes beyond that needed for the stabilization of active disease and includes restorative treatment designed

primarily to enhance esthetics (eg, ceramic veneers) and to provide optimum function (eg, replacement of missing teeth using FPDs) and resistance to oral stresses (eg, cast restorations).

One of the primary benefits of segregating the restorative treatment into these categories is that a formal reevaluation is completed at the end of the disease control phase before progressing into the definitive treatment phase. This is not to imply that the phased approach to restorative care is so structured as to preclude consideration of definitive care in the disease control phase. For example, a patient who has demonstrated good plaque control and has adequate finances may require an amalgam build-up on a maxillary first premolar as part of the disease control phase of treatment. Ultimately, it is anticipated that this tooth will receive an all-ceramic crown. However, this amalgam build-up would be esthetically unacceptable to the patient, and so the amalgam core may be immediately prepared for a crown and a provisional restoration placed during the disease control phase of care. The phased approach incorporates into the plan the opportunity to modify or curtail restorative treatment after the control of caries and the replacement of defective restorations. There can be many reasons for altering the original treatment plan, including the patient's desires, disease risk, failure to accomplish disease control, finances, or the doctor-patient relationship.

The patient's financial situation or third-party payment guidelines may dictate that treatment be divided into stages and completed over a period of time. Organization of treatment into phases serves the patient's most urgent needs first, directing resources into the management of active disease and allowing less acute problems to be addressed as finances permit.

As previously emphasized, treatment planning for restorative dentistry requires that the dentist recognize the sequence in which restorative care should be provided within the context of the overall plan. It is not enough to be able to envision the final goal of treatment; one must be able to visualize each step that must be accomplished to achieve this goal. The following example illustrates this point.

A patient presented stating that he wished to "close the spaces" between his front teeth (Fig 2-24a). Upon evaluation, this seemingly simple request revealed a complex set of problems. The dentist recognized the problem associated with the patient's chief complaint: diastemata resulting from a tooth-size versus jaw-size discrepancy. The dentist also recognized other esthetic problems associated with the anterior teeth (Fig 2-24b): fluorosis-related discolorations of the teeth and incomplete exposure of the crowns of the anterior teeth due to altered passive eruption of the maxillary lateral incisors.

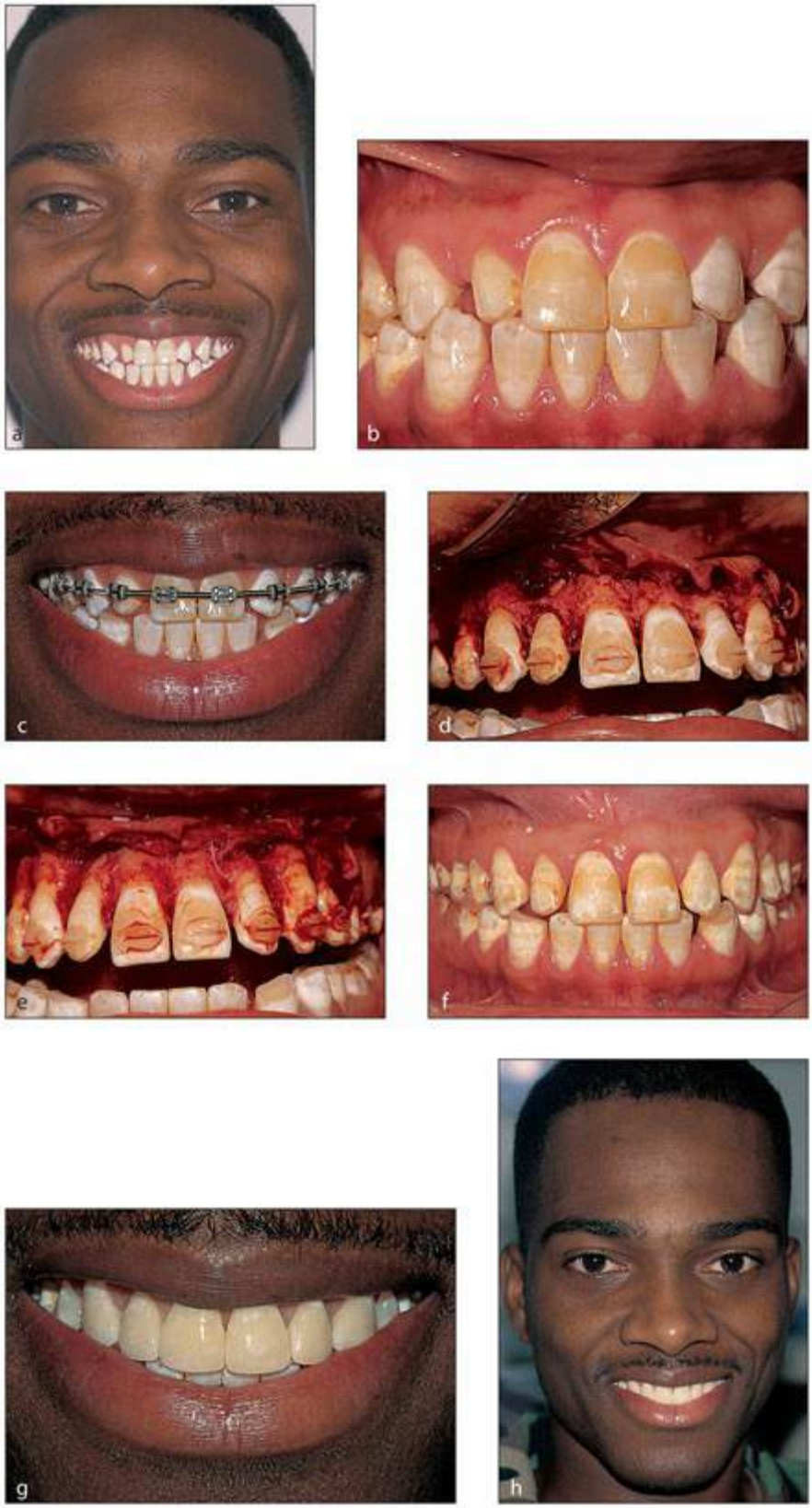


Fig 2-24*(a)* The patient wished to close the diastemata adjacent to the maxillary lateral incisors. *(b)* A close-up view reveals, in addition to the diastemata and discolored anterior teeth, the unesthetically short clinical crowns of the maxillary lateral incisors. For better space distribution prior to ceramic veneer fabrication, the maxillary lateral incisors need to be repositioned distally

and their crowns lengthened. The mandibular canines are obstructing movement of the maxillary lateral incisors into the desired locations. *(c)* The maxillary lateral incisors have been repositioned orthodontically. Space was created by odontoplasty of the mandibular canines followed by resin composite restorations. The space redistribution permits diastema closure to be completed by adding restorative material to all six anterior teeth, which avoids the problem of making any single tooth excessively wide. *(d)* Mucogingival flap elevation reveals the osseous crest to be immediately adjacent to the cementoenamel junctions of the maxillary lateral incisors. This anatomical relationship is responsible for the gingiva covering a portion of the crowns of these teeth. *(e)* Ostectomy and osteoplasty have created approximately 3 mm of space for the combination of sulcus depth, connective tissue attachment, and epithelial attachment. This space will allow the gingival crest to reside at the level of the cervical margin of the veneers, displaying the entire crown of each tooth. *(f)* Three months after surgery, healing is complete and the teeth are ready for veneer preparation. *(g)* One month after veneer placement (maxillary first premolar to first premolar). The spaces have been closed, and the fluorosis-related discoloration has been eliminated. *(h)* The patient was extremely satisfied with the final results of his multidisciplinary treatment.

The dentist considered possible solutions to the diastema problem. The two most common solutions to this type of space-related problem are *(1)* closure of the spaces by the placement of restorations and *(2)* orthodontic retraction of the maxillary teeth, creating a smaller arch perimeter and reducing or eliminating the spaces between the teeth. An occlusal analysis revealed that the maxillomandibular dental relationships would not permit retraction of the maxillary anterior teeth (see Fig 2-24b). Thus, complete space closure would require filling all of the open spaces with tooth-colored restorative materials. A space analysis and a diagnostic wax-up revealed that closure of all of the spaces would result in excessive (unesthetic) widening of the maxillary lateral incisors and canines. Complete space closure would be esthetically acceptable only if it were accomplished by adding a small amount of restorative material to all of the maxillary anterior teeth. The two options available were *(1)* partial closure of the diastemata with tooth-colored restorative material, leaving small spaces between the maxillary lateral incisors and canines, or *(2)* orthodontic redistribution of the existing spaces followed by complete space closure using tooth-colored restorations placed on all six maxillary anterior teeth.

When presented with these possibilities, the patient stated that he would prefer complete space closure and would be willing to undergo orthodontic treatment to accomplish this. The dentist then visualized the optimal treatment goal and realized that the maxillary lateral incisors would need to be moved to a more distal position to equalize the anterior spacing. This was added to the problem list. Visualizing the distal movement of the maxillary lateral incisors, the dentist realized that the positions of the mandibular canines would interfere with this movement. This presented a new problem, which was added to the problem list. The dentist

considered two options: *(1)* orthodontic movement of the mandibular canines or *(2)* alteration of the contour of the mandibular canines to accommodate the repositioning of the maxillary lateral incisors. The orthodontic movement required to reposition the mandibular canines was deemed unfeasible, so the second option was selected.

Having determined the feasibility of orthodontic space redistribution for the maxillary anterior teeth, the dentist visualized the final result. Increasing the width of the anterior teeth using tooth-colored restorations (ceramic veneers) would increase the tooth-width to tooth-height ratio, making the maxillary lateral incisors appear unesthetically short and wide. Diagnostic periodontal probing and bone sounding procedures were completed to address this newfound problem. The relative locations of the cementoenamel junctions and the distances from the gingival crests to the osseous crests of the maxillary lateral incisors were determined. It was decided that the ideal solution for the "short tooth" problem of the maxillary lateral incisors was to expose the complete crown of these teeth through surgical crown lengthening before ceramic veneer placement. This plan was presented and was accepted by the patient.

In stepwise fashion, the entire problem complex was broken down into its individual components. Each component and its proposed solution were assessed. Any new problems that were created by proposed treatment were considered. The final chain of treatment was established and presented to the patient. By recognizing which form of treatment was required to address each component problem, the dentist was able to plan the entire sequence before initiating treatment. All of the proposed procedures were completed, and the treatment of the patient's anterior esthetic problem was realized (Figs 2-24c to 2-24h).

The Dental Record

Accurate and descriptive record keeping is essential to quality dental care. The dental chart should include findings from the history and examination, the problem list, the treatment plan, and a description of the treatment accomplished. This record serves several purposes:

- Organization and documentation of the examination findings, the problem list, the treatment plan, and the treatment rendered
- Documentation for third-party payment, if applicable
- Legal purposes

- Forensic purposes

Organizing and documenting the examination findings and the problem list enable the dentist to evaluate the patient's dental problems and plan the treatment when the patient is no longer present. Once treatment has begun, documentation of the sequenced treatment plan also permits the dentist to review the anticipated treatment without the need to reconsider the entire treatment planning process. Dental records should include the following information:

- Charting of examination findings, including existing restorations and dental relationships (eg, diastemata, dentoalveolar extrusion, tilted teeth), existing periodontal and endodontic conditions, occlusal relationships, and caries lesions and defective restorations
- Medical history and consultations
- Problem list
- Treatment plan
- Description of treatment provided
- Informed consent documentation
- Follow-up assessment

In addition to handwritten or, more recently, electronic entries in a dental record, pictorial charting is an efficient means of recording a great deal of information in a small area (Figs 2-25 and 2-26). Intraoral imaging devices, either conventional cameras or videographic recording devices, provide an extremely effective means of recording findings and documenting treatment and are an ideal complement to a pictorial charting system. These offer the added advantage of simplifying communication with the patient and third-party funding agencies.

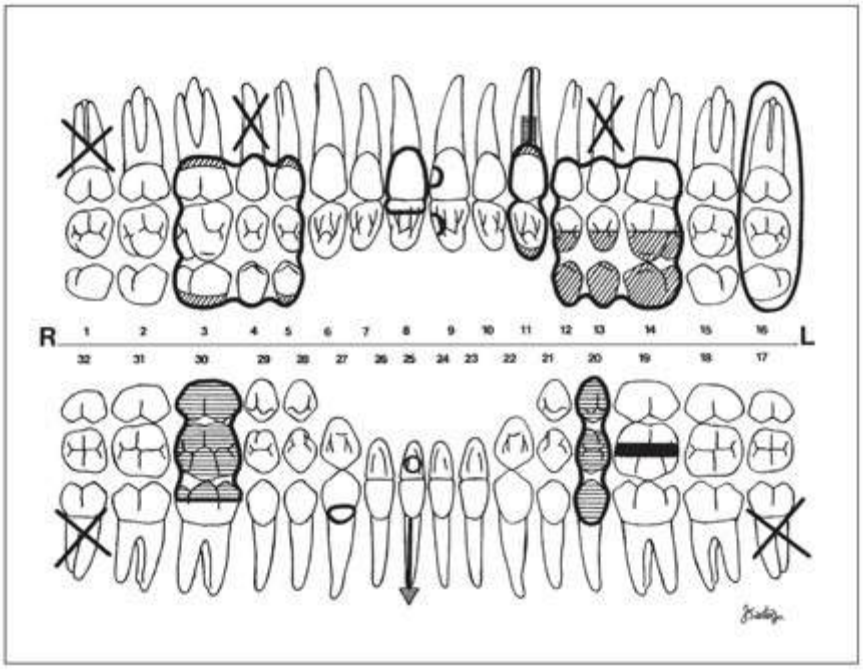
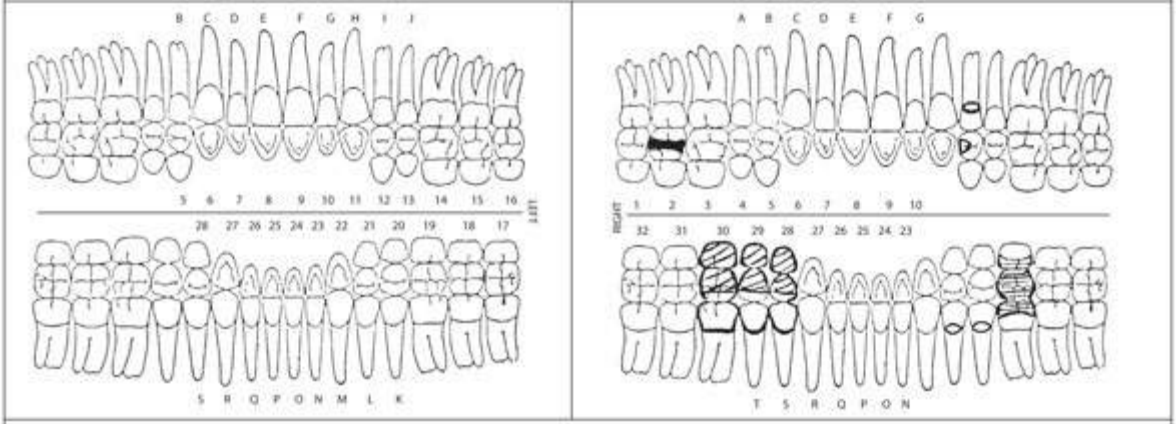


Fig 2-25 Example of a pictorial charting system used to record dental restorations. Any system that distinguishes among the various restorations is acceptable. In this example, tooth 1 is missing; tooth 4 has been replaced with a metal-ceramic FPD that extends from tooth 3 to tooth 5 with ceramic occlusal coverage; tooth 8 has a facial veneer; tooth 9 has a mesial resin composite restoration; tooth 11 has been endodontically treated and has a post and metal-ceramic crown; tooth 13 has been replaced by a metal-ceramic FPD that extends from tooth 12 to tooth 14 with metal occlusal coverage; tooth 16 is impacted; tooth 17 is missing; tooth 19 has an MOD amalgam restoration; tooth 20 has been restored with a metal crown; tooth 25 has been endodontically treated, received a retrograde restoration, and has a resin composite restoration in the lingual access opening; tooth 27 has a facial tooth-colored restoration; tooth 30 has a metal three-quarter crown; and tooth 32 is missing.

PATIENT: Blank, Felina D.

TREATMENT PLAN BY APPOINTMENT



REMARKS

Appt	Problem	Treatment	Time	Total cost	Patient cost	Insur. code	Comp. date
1	CC: #19 Hypertension	Discuss & defer until after med Tx Refer to Dr Dogood	30 min				
2	Tx plan Caries Periodontics	Present Tx plan Counseling; diet and hygiene instruction Plaque index; prophylaxis; scale/root plane mandibular posterior sextants	Dr: 30 min Hygiene: 60 min				
3	Cracked: #19	Prep gold onlay	45 min				
4	Defective restorations: #2, #12	#2: MOD amalgam #12: MO resin composite #12: Class 5 resin composite	60 min				
5	Caries: #20, #21 #19	Resin-modified glass ionomer, Class 5 facial #20, #21 Deliver #19 gold onlay	60 min				
6	Biologic width: #3	Crown lengthening surgery	45 min				
7	Biologic width: #3	Postop evaluation (1 week)	10 min				
8	Reevaluation	Perio and hygiene reeval; confirm definite phase plan	20 min				
9	Missing: #29 Supraerupted: #4	Prep FPD, #28–#30 Shorten	2 h				
10	Missing: #29	Deliver FPD, #28–#30 Impression for occlusal splint	90 min				
11	Wear: #6–#11	Deliver occlusal splint	30 min				
12	Fluorosis: #8	Microabrasion; maxillary impression for bleaching stent	30 min				
13	Color: maxillary anterior teeth	Deliver bleaching stent & give use instructions	15 min				
14	Color: maxillary anterior teeth	Reevaluate bleaching results; reinforce hygiene	10 min				
15	Maintenance	Prophylaxis; fluoride; reexamination	Dr: 10 min Hygiene: 30 min				

Fig 2-26 Example of a combined written and pictorial treatment planning worksheet. The pictorial charts are for use in recording both completed treatment *(left)* and treatment yet to be accomplished *(right)*. Once completed, the treatment charted in pencil on the right may be erased. The remarks section is for making comments, generally in pencil (erased and updated as needed). The area below the remarks section is for the sequenced treatment plan. A treatment planning sheet such as this allows for quick review of the overall treatment plan and provides a profile of the current status of treatment.

The format used to document the care provided should reflect an orderly and

logical diagnostic and treatment sequence. A method commonly used in medicine that satisfies this requirement is the SOAP format. *SOAP* is an acronym for the steps involved whenever any treatment is rendered. *S* refers to subjective findings. This includes a summary of the patient's medical status, chief complaint, a description of his or her symptoms, and any other relevant information the patient provides. The patient's own words should be used as much as possible. The *O* refers to objective findings. These include both extraoral and intraoral examination findings (including vital signs) and the results of consultations and diagnostic tests. *A* refers to the assessment, which is the dentist's diagnosis, based on the subjective and objective findings present as well as overall patient assessment (for example, a patient with a high caries risk). The *P* refers to the plan of treatment (when the treatment is rendered immediately, it refers to the procedures or the treatment itself). An example of the use of the SOAP format is provided in Box 2-3.

Box 2-3 **Example of SOAP format**

S: Patient presents complaint of a "toothache that began yesterday and has hurt all night." (Patient points to tooth 30.) Patient states that ice water reduces the pain.

O: Teeth 27 and 30 are within normal limits (WNL) to percussion, palpation, and periodontal examination and are vital and normal to cold testing. Tooth 31 is painful to percussion, and pain is alleviated with the application of cold. Radiographs of teeth 27 to 31 are unremarkable, except for a deep mesio-occlusal amalgam restoration on tooth 31

A: Tooth 31 has irreversible pulpitis.

P: Patient advised of diagnosis. Patient consents to endodontic treatment on tooth 31. Appointment for pulpectomy on tooth 31 at 3:00 PM today.

The SOAP format is an excellent guide in the performance and documentation of care when a challenging diagnostic problem presents itself; however, it is less suited to the routine restorative care provided based on the treatment plan. When a straightforward diagnosis is made in the absence of symptoms and patient complaint, a more concise form of documentation is appropriate (Box 2-4).

Box 2-4 **Example of concise form**

DX: Caries in tooth 3, vital to cold and asymptomatic.

TX: Tooth 3: MOD amalgam (Tytin), Amalgambond, rubber dam. Local anesthesia: 36 mg lidocaine, with 0.018 mg epinephrine.
Plan: Schedule appointment for preparation of veneers for teeth 5 to 12.

There are times when the identification of a deceased individual must be accomplished through the use of dental records. A complete record of the dentition and restorations, a radiographic survey, and photographic records are useful for identification purposes.

The dental record is a legal document. The nature and clarity of the entries made should reflect the knowledge that it may be needed in a court of law to document examination findings, informed consent, and treatment completed. The records should be accurate and should contain the elements listed above. They should not contain erasures or text that has been obliterated by any means. If errors are made, a single line should be drawn through the mistake and the change should be initialed and dated. In the retrospective review of a legal investigation, the descriptiveness and clarity of the record is often held to be an indication of the quality of care provided.

Summary

Treatment planning for restorative dentistry can be a complex undertaking. Use of a logical and orderly problem-oriented approach can simplify the process. The following principles have been offered as guidelines:

- Be aware of pathoses that may be encountered and be able to distinguish the normal from the abnormal and stable from risk-prone situations.
- Organize abnormal findings into a problem list.
- Envision an overall restorative goal for the patient. This is the anticipated final state of rehabilitation. Not every patient can be restored to the ideal, but each patient has an optimum state of health that can be obtained given his or her circumstances.
- Determine a treatment plan for each problem so that each treatment contributes to the achievement of the anticipated ultimate treatment goal. This is the linchpin of restorative treatment planning. It requires that the dentist consider a number of

factors before selecting the optimum restorative option. Chief among these considerations are the overall goal of treatment, the functional and esthetic needs of each restorative situation (using the existing dentition and restorations as an indicator of the performance of future restorations), the strengths and weaknesses of the various restorative materials available, and the amount and location of remaining tooth structure.

- Recognize the sequence of steps needed to achieve a specific restorative objective. The dentist must know, for example, that a tooth fractured at the level of the osseous crest and in need of a crown will require endodontic treatment, a post and core, and crown lengthening surgery before fabrication of the definitive restoration.
- Sequence the treatment based on a logical model. Control active disease processes first, beginning treatment with teeth in the most acute need of care. Complete as much care as is feasible in each sextant at the same appointment. Establish a restorative prognosis for key teeth early in the treatment schedule. Consider nondental factors (especially third-party payment guidelines and time-related limits) when planning the treatment schedule.

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